Project Report On

Koode - A Smart Companion for Alzheimer's Patients

Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology

in

Computer Science and Engineering

By

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March 2026

CERTIFICATE

*This is to certify that the project report entitled **"Koode - A Smart Companion for Alzheimer's Patients"** is a bonafide record of the work done by **Christopher Cleatus (U2203071)**, submitted to the Rajagiri School of Engineering & Technology (RSET) (Autonomous) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B. Tech.) in Computer Science and Engineering during the academic year 2025-2026.*

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ACKNOWLEDGMENT

I wish to express my sincere gratitude towards **Rev. Dr. Jaison Paul Mulerrikal CMI**, Principal of RSET, and **Dr. Preetha K G**, Head of the Department of Computer Science and Engineering, for providing me with the opportunity to undertake my project, "Koode - A Smart Companion for Alzheimer's Patients".

I am highly indebted to my project coordinators, **Dr. Jisha G**, Associate Professor, Department of Computer Science and Engineering, and **Dr. Preetha K G**, Professor, Department of Computer Science and Engineering, for their valuable support.

It is indeed my pleasure and a moment of satisfaction for me to express my sincere gratitude to my project guide **Mr. Sebin Jose** for his patience and the priceless advice and wisdom he has shared with me.

Last but not the least, I would like to express my sincere gratitude towards all other teachers and friends for their continuous support and constructive ideas.

Christopher Cleatus

Abstract

Alzheimer's disease and other forms of dementia affect a person's memory, orientation, and ability to perform daily activities independently. As the condition progresses, patients may forget familiar people, wander away from safe locations, or face difficulty seeking help during emergencies. Continuous monitoring by caregivers is often required, which can be challenging to maintain.

This project presents the design and development of a smart wearable system to assist individuals in the early stages of Alzheimer's disease while supporting caregivers. The device integrates technologies such as voice recognition, fall detection using motion sensors, GPS-based location tracking, and geo-fencing alerts. Sensor data collected through an embedded microcontroller is transmitted to a connected mobile application and backend server for processing and alert generation.

The system detects abnormal events such as falls and safe-zone breaches and automatically notifies caregivers. The caregiver application provides real-time alerts, location updates, and activity logs for continuous monitoring. The proposed solution aims to enhance patient safety, improve independence, and reduce caregiver burden through an intelligent and reliable wearable monitoring system.

Contents

List of Abbreviations

AI - Artificial Intelligence

ANN - Artificial Neural Network

BLE - Bluetooth Low Energy

CNN - Convolutional Neural Network

EER - Equal Error Rate

ESP32 - System on a Chip Microcontroller

FCM - Firebase Cloud Messaging

GE2E - Generalized End-to-End

GPS - Global Positioning System

IDE - Integrated Development Environment

IoT - Internet of Things

LSTM - Long Short-Term Memory

MEMS - Micro-Electro-Mechanical Systems

ML - Machine Learning

MPU6050 - Accelerometer and Gyroscope Sensor

MQTT - Message Queuing Telemetry Transport

RF - Random Forest

SDK - Software Development Kit

SVM - Support Vector Machine

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Chapter 1

Introduction

Koode is an assistive technology platform designed to support elderly people and patients who require continuous monitoring, especially those with conditions like Alzheimer's or memory-related disorders. The system integrates wearable devices, mobile applications, and intelligent backend services to ensure safety, monitoring, and timely assistance for the user.

The main goal of the project is to create a smart companion system that helps caregivers monitor patients in real time while allowing patients to live more independently.

1.1 Background

In recent years, there has been a growing need for technological solutions that assist elderly individuals and patients who require continuous monitoring. With the increase in age-related cognitive disorders such as Alzheimer's disease and other memory impairments, many individuals face difficulties in performing daily activities independently. These conditions often lead to situations where patients may forget important tasks, wander away from safe locations, or experience accidents such as falls. As a result, caregivers and family members must constantly monitor the well-being and safety of these individuals.

Traditional caregiving methods rely heavily on manual supervision, which can be physically and emotionally demanding for caregivers. Continuous in-person monitoring is not always feasible, especially when caregivers have other responsibilities or when patients prefer to maintain a certain level of independence. In many cases, the lack of timely assistance during emergencies can lead to serious consequences for the patient's health and safety.

Advancements in technologies such as the Internet of Things (IoT), wearable devices, and mobile applications have created new opportunities to address these challenges. Wearable sensors can collect real-time information about a user's activities, location, and movement, while mobile and cloud technologies enable this information to be transmitted and monitored remotely. These technological developments allow caregivers to stay informed about the patient's condition without the need for constant physical presence.

The Koode system is developed in response to these needs, aiming to provide a smart monitoring and assistance platform for individuals who require continuous supervision. By integrating wearable sensing devices, real-time data communication, and mobile-based monitoring, the system enables caregivers to track important events such as location updates and fall incidents. Through timely alerts and accessible information, the platform helps improve patient safety while reducing the burden on caregivers. In this way, Koode contributes to the development of intelligent assistive solutions that support independent living while ensuring reliable monitoring and care.

1.2 Problem Definition

Elderly individuals and patients with cognitive or mobility-related conditions often require continuous monitoring to ensure their safety, which can be challenging for caregivers to manage at all times. The aim of this project is to develop an assistive monitoring system that uses wearable devices and mobile technology to track patient activity, detect critical events such as falls, and provide real-time alerts to caregivers.

1.3 Scope and Motivation

Scope: The Koode project aims to develop an assistive monitoring system that supports elderly individuals and patients who require continuous supervision. The system focuses on integrating wearable devices with mobile and cloud-based technologies to monitor important parameters such as location tracking, fall detection, and voice-based identification. It enables caregivers to receive real-time updates and alerts through a connected mobile application, allowing them to respond quickly in emergency situations. The platform also maintains logs of events and activities, providing useful information for caregivers to understand the patient's daily patterns. By combining sensing technologies, real-time communication, and mobile interfaces, the project aims to create a reliable and user-friendly monitoring solution.

Motivation: The motivation for developing the Koode system arises from the growing challenges faced by caregivers in continuously monitoring elderly individuals and patients with cognitive or mobility-related conditions. Traditional caregiving methods often require constant physical supervision, which can be demanding and sometimes impractical. Delays in detecting emergencies such as falls or wandering can lead to serious health and safety risks. Advances in wearable technology, IoT, and mobile communication offer opportunities to create systems that can monitor users remotely and provide timely alerts when abnormal situations occur. Therefore, this project is motivated by the need to enhance patient safety, reduce caregiver burden, and promote a more independent lifestyle for individuals who require continuous care.

1.4 Objectives

1. To develop an assistive monitoring system that improves the safety and well-being of elderly individuals and patients who require continuous supervision.
2. To design and implement a wearable-based solution capable of monitoring user location and detecting fall events in real time.
3. To enable caregivers to remotely monitor patient activities and receive instant alerts through a connected mobile application.
4. To integrate voice identification techniques for recognizing individuals interacting with the patient.
5. To store and manage activity logs and event data using a secure cloud-based backend system.
6. To create a reliable and user-friendly platform that supports independent living while reducing the burden on caregivers.

1.5 Challenges

Developing the Koode assistive monitoring system involves several technical and practical challenges:

- **Reliable Sensor Monitoring** Ensuring accurate detection of events such as falls and user movement using wearable sensors without generating false alerts.

- **Real-Time Data Communication** Maintaining stable communication between the wearable device, backend server, and mobile application to provide timely updates and alerts.

- **Voice Identification Accuracy** Implementing a reliable voice recognition system that can correctly identify individuals despite variations in voice samples or background noise.

- **System Integration** Integrating multiple technologies including wearable hardware, cloud services, and mobile applications into a single seamless platform.

- **Data Security and Reliability** Ensuring that sensitive user data is securely stored and that the system consistently delivers accurate monitoring and notifications.

1.6 Assumptions

1. **Users will consistently wear the monitoring device** It is assumed that the patient or user will wear the wearable device regularly so that the system can continuously monitor their activities and detect events such as falls.

2. **Stable internet connectivity is available** The system assumes that the wearable device and mobile application have access to a stable internet connection to transmit data and receive real-time alerts.

3. **Caregivers actively monitor the application** It is assumed that caregivers or family members will regularly use the mobile application to monitor updates and respond to alerts generated by the system.

4. **Sensor readings are reasonably accurate** The project assumes that the sensors used in the wearable device can reliably capture movement and activity data required for fall detection and monitoring.

5. **Authorized users access the system** It is assumed that only authorized caregivers or family members will access the monitoring application and view the user's

data.

1.7 Societal / Industrial Relevance

Industrial Relevance

- **Advancement in Healthcare Monitoring** – The system contributes to the development of smart healthcare solutions by integrating wearable devices, IoT technologies, and mobile applications for patient monitoring.
- **Remote Patient Monitoring Systems** – Healthcare providers and organizations can use such systems to monitor patients remotely, reducing the need for constant in-person supervision.
- **Integration with Smart Healthcare Platforms** – The project demonstrates how wearable sensors, cloud services, and mobile applications can be combined to build scalable healthcare monitoring systems.
- **Support for Assisted Living Technologies** – The system can be applied in assisted living facilities and elderly care centers to improve patient safety and monitoring efficiency.

Societal Relevance

- **Improved Safety for Elderly Individuals** – The system helps monitor elderly people and patients with cognitive conditions, enabling faster response during emergencies such as falls.
- **Reduced Caregiver Burden** – Caregivers and family members can monitor patients remotely, reducing the stress and effort required for constant supervision.
- **Encouraging Independent Living** – By providing continuous monitoring and timely alerts, the system allows elderly individuals to live more independently while still being supported.
- **Better Quality of Life** – Early detection of risky situations and quick assistance can improve the overall well-being and safety of patients.

1.8 Summary of Chapter

This chapter introduced the Koode project and discussed the background and motivation behind developing an assistive monitoring system for individuals requiring continuous supervision. The problem definition, objectives, and scope of the project were presented to outline the purpose and goals of the system. The chapter also highlighted the key challenges involved in developing such a system and the assumptions considered during its design. In addition, the societal and industrial relevance of the project was discussed, emphasizing its potential benefits in healthcare monitoring and assisted living technologies. Overall, this chapter provides a foundational understanding of the project and its significance.

Chapter 2

Literature Survey

2.1 Introduction

With the increasing global prevalence of Alzheimer's disease and the growing aging population, ensuring continuous care and safety for affected individuals has become a critical challenge. Traditional caregiving methods are often insufficient for providing real-time support and monitoring, especially when patients are alone. Wearable IoT technology, combined with advanced sensor systems and AI, offers promising solutions to these challenges by enabling real-time health monitoring, fall detection, location tracking, and interactive assistance.

This chapter reviews key research and technological advancements in wearable fall detection, voice recognition, and remote patient monitoring systems specifically designed for elderly and Alzheimer's patients. It compares different methodologies, highlights the benefits and limitations identified in existing works, and identifies the gaps that this project aims to address with an integrated, secure, and user-friendly IoT companion system for Alzheimer's care.

2.2 Survey Papers

2.2.1 Implementing GPS Trackers for People with Dementia at Risk of Wandering

This study investigates the use of GPS trackers to reduce wandering risks in dementia patients. Dementia often leads to patients wandering unsafely, causing harm and stress to both patients and caregivers. The research tests the effectiveness of wearable GPS devices that provide real-time location tracking and alerts through caregiver monitoring apps.

Objective: To evaluate whether GPS tracking devices can improve safety by timely alerting caregivers when dementia patients move outside predefined safe zones and to assess patient independence and caregiver satisfaction.

Key Features:

- Real-time GPS tracking of patients' location.
- Geofencing to set safe zones and trigger alerts when breached.
- Fall detection capability.
- Two-way communication via SOS button and speaker.
- Caregiver app for monitoring and alert management.

Advantages:

- Improved patient safety through timely alerts and fast retrieval.
- Increased patient independence with monitored freedom of movement.
- Reduced caregiver burden and enhanced peace of mind.
- Positive caregiver feedback regarding usability and confidence in home care.

Disadvantages and Limitations:

- Most participants had moderate to severe dementia and could not self-report experiences.
- Lack of a control group makes it difficult to isolate GPS tracking's direct impact.
- Potential dependency on stable GPS signals and network connectivity.

Relevance to Project: This study supports the inclusion of GPS tracking with geofencing in the Alzheimer's Companion IoT system. It provides evidence that real-time location monitoring enhances patient safety and independence while reducing caregiver stress, directly aligning with the project's goal to provide secure, intelligent assistance for Alzheimer's patients at risk of wandering.

2.2.2 CareFall – Automatic Fall Detection through Wearable Devices and AI Methods

CareFall presents an automatic fall detection system using wearable devices equipped with sensors like 3-axis accelerometers and gyroscopes. The system continuously monitors user movements to detect falls, differentiating them from daily activities by analyzing sensor data with AI methods.

Objective: To develop an accurate, real-time fall detection system that minimizes false alarms and provides timely emergency alerts to caregivers or emergency services, enhancing elderly and dementia patient safety.

Key Features:

- Utilizes wearable smartwatches with integrated accelerometer and gyroscope sensors.
- Employs threshold-based and machine learning classification techniques (SVM, RF, ANN).
- Extracts statistical features from sensor signals for robust fall detection.
- Provides automatic, hands-free fall alerts and emergency notifications.

Advantages:

- High accuracy in detecting falls using sensor fusion and machine learning.
- User-friendly and deployable on commercial smartwatches.
- Real-time detection enables faster emergency responses.

Disadvantages and Limitations:

- Threshold-based methods may result in false alarms.
- Performance heavily depends on sensor placement.
- Requires large labeled datasets for effective training.
- Smartwatch battery and connectivity limitations may affect continuous monitoring.

Relevance to Project: This research justifies integrating AI-based fall detection methods combining accelerometer and gyroscope data into the Alzheimer's Companion IoT system, ensuring timely alerts and improved safety.

2.2.3 Generalized End-to-End Loss for Speaker Verification

This paper introduces a novel loss function called Generalized End-to-End (GE2E) loss designed to improve the efficiency and accuracy of speaker verification systems.

Objective: To develop a loss function that accelerates training and enhances speaker verification performance by improving discrimination between speakers and reducing error rates.

Key Features:

- Uses centroid-based comparisons of speaker embeddings.
- Softmax-based formulation to improve speaker discrimination.
- MultiReader technique for domain adaptation across different dialects.

Advantages:

- Reduces speaker verification Equal Error Rate (EER).
- Decreases training time significantly.
- Simplifies model training without example selection.

Disadvantages and Limitations:

- Designed mainly for speaker verification tasks.
- Requires large training datasets.

Relevance to Project: This research is relevant to the voice recognition module of the Alzheimer's Companion IoT system, enabling reliable identification of known voices for better patient-caregiver interaction.

2.2.4 Secure IoT Assistant-Based System for Alzheimer's Disease

This paper presents a secure IoT assistant system designed to support Alzheimer's patients by providing real-time monitoring, safety, and communication features.

Objective: To develop an intelligent, secure, and user-friendly IoT system that continuously monitors Alzheimer's patients and improves their quality of life.

Key Features:

- Voice recognition distinguishing family members from strangers using CNN models.
- Real-time GPS tracking with geofencing alerts.
- Fall detection using accelerometer and gyroscope sensors.
- Secure cloud synchronization for data storage.
- Caregiver mobile application for monitoring and alerts.

Advantages:

- Provides 24/7 monitoring for patient safety.
- Voice interaction reduces social isolation.
- Real-time alerts improve caregiver response time.
- Uses affordable IoT hardware.

Disadvantages and Limitations:

- Device dependence if patients forget to wear it.
- Possible false positives in detection systems.
- Battery life constraints.
- GPS and internet connectivity issues.

Relevance to Project: This research provides foundational methodologies for the Alzheimer's Companion system, validating the integration of voice recognition, fall detection, and GPS tracking into a comprehensive IoT healthcare platform.

2.3 Comparison of Survey Papers

Table 2.1: Comparison of Key Research Papers

Paper Title	**Methodology**	**Strengths**	**Limitations**
Implementing GPS Trackers for People with Dementia at Risk of Wandering	Uses wearable GPS devices, geofencing alerts, and caregiver app monitoring to track patient location in real time.	Timely alerts improve patient safety; increases independence; reduces caregiver stress.	Dependent on stable GPS/network; may not generalize across populations.
CareFall – Automatic Fall Detection through Wearable Devices and AI Methods	Employs smartwatches with accelerometer and gyroscope sensors; uses threshold-based and machine learning classification for fall detection.	High accuracy; real-time alerting; simple deployment on commercial devices.	Threshold methods can trigger false alarms; battery and connectivity limitations; requires labeled datasets.
Generalized End-to-End Loss for Speaker Verification	Uses centroid-based speaker embeddings and end-to-end softmax loss for training speaker verification models.	Reduced error rate; faster training; suitable for voice identification in wearable systems.	Requires large speaker datasets; mainly focused on speaker verification tasks.
Secure IoT Assistant-Based System for Alzheimer's Disease	Integrates AI-based voice recognition, GPS geofencing, sensor-based fall detection, and encrypted cloud storage for patient monitoring.	Unified monitoring system; improves patient safety and independence; real-time alerts.	Device dependence; risk of false alarms; battery and GPS limitations.

2.4 Gaps Identified

The literature survey reveals that while individual technologies such as wearable fall detection systems, voice-based authentication, and GPS tracking with geofencing have demonstrated effectiveness in isolation, there is a clear gap in integrating these functionalities into a single cohesive platform specifically designed for Alzheimer's care. Most existing solutions focus on only one aspect of monitoring or assistance, such as fall detection or location tracking, without providing a comprehensive system that manages safety, communication, and real-time caregiver alerts simultaneously.

The primary gap identified is the absence of a unified IoT-based solution that combines sensor-driven fall detection, continuous GPS monitoring, voice recognition, and secure cloud-based data management tailored to the needs of Alzheimer's and dementia patients and their caregivers. Existing research also lacks extensive evaluation of how such a multi-module system can improve patient independence, safety, and overall quality of life in real-world environments. The current project aims to address this gap by developing an integrated, accessible, and secure smart companion platform for Alzheimer's care.

2.5 Summary of the Chapter

This chapter presented a comprehensive review of research works and technologies related to the development of the Alzheimer's Companion IoT system. It examined key studies in areas such as GPS-based patient tracking, wearable fall detection using sensors and artificial intelligence, voice recognition for speaker identification, and secure IoT systems for real-time health monitoring. Each study highlighted important methodologies, strengths, and limitations that contribute to the understanding of intelligent patient-support systems.

The literature analysis revealed a significant gap in the integration of these technologies into a unified platform designed specifically for Alzheimer's care. While existing solutions effectively address individual aspects such as fall detection or location monitoring, they lack a comprehensive system that ensures continuous monitoring, communication, and safety. The next chapter will present the proposed system design that aims to bridge this gap by integrating these technologies into a single IoT-based assistant that enhances patient safety, caregiver awareness, and independent living.

Chapter 3

System Design

In the Koode project, the system design integrates wearable hardware, sensor data processing, and a mobile application to enable real-time monitoring and alert generation. The system collects data from sensors, processes it, and sends notifications to users when necessary.

This chapter presents the overall architecture of the system, component design, module division, tools and technologies used, project timeline, and key deliverables involved in the development of the Koode system.

3.1 Overall Architecture

The overall architecture of the Koode system is designed to integrate a wearable hardware device with a mobile application and cloud-based backend services to enable real-time monitoring and alert communication. The system follows a layered architecture where data flows from the sensing device to the user interface through processing and communication layers.

At the hardware level, a wearable device equipped with sensors continuously monitors the user's physical movements and environmental data. The sensors capture information such as acceleration and motion patterns, which are used to detect abnormal events such as falls or sudden impacts.

The collected sensor data is processed by a microcontroller within the wearable device. The microcontroller analyzes the sensor readings and determines whether the data indicates a potential emergency situation. When such an event is detected, the device prepares the data for transmission.

The processed information is then transmitted to the cloud infrastructure through a network communication module. The backend system receives and stores the data while also triggering the appropriate notification services. The backend also manages user authentication, event records, and communication between devices and applications.

On the user side, a mobile application acts as the primary interface for monitoring and receiving alerts. The application displays notifications, location information, and emergency alerts generated by the system. Caregivers or registered contacts can receive these alerts in real time, allowing them to respond quickly when necessary.

Through this architecture, the Koode system ensures seamless communication between hardware, cloud services, and the user interface. The modular structure also allows the system to be expanded in the future by adding new sensors, improved detection algorithms, or additional notification services.

3.2 Component Design

3.2.1 Architecture Diagram

The architecture diagram illustrates the major components of the Koode system and the flow of information between them. The system consists of a wearable device equipped with motion sensors, a communication module, a backend server, and a mobile application used by caregivers or users.

The wearable device continuously collects motion data using sensors such as an accelerometer and gyroscope. The microcontroller processes this data to detect abnormal events such as falls. When a potential emergency is detected, the device transmits the information to the backend system through a network communication module.

The backend server manages data storage, event processing, and notification services. It records device data in the database and triggers alerts when emergency events are identified. The backend also supports user authentication and manages communication between the device and the mobile application.

The mobile application provides the user interface for monitoring the system. It displays alerts, location information, and system status updates. Caregivers receive real-time notifications through the application, allowing them to respond quickly during emergencies.

The architecture diagram therefore demonstrates how data flows from the wearable device to the backend infrastructure and finally to the user interface, ensuring efficient monitoring and rapid alert communication.

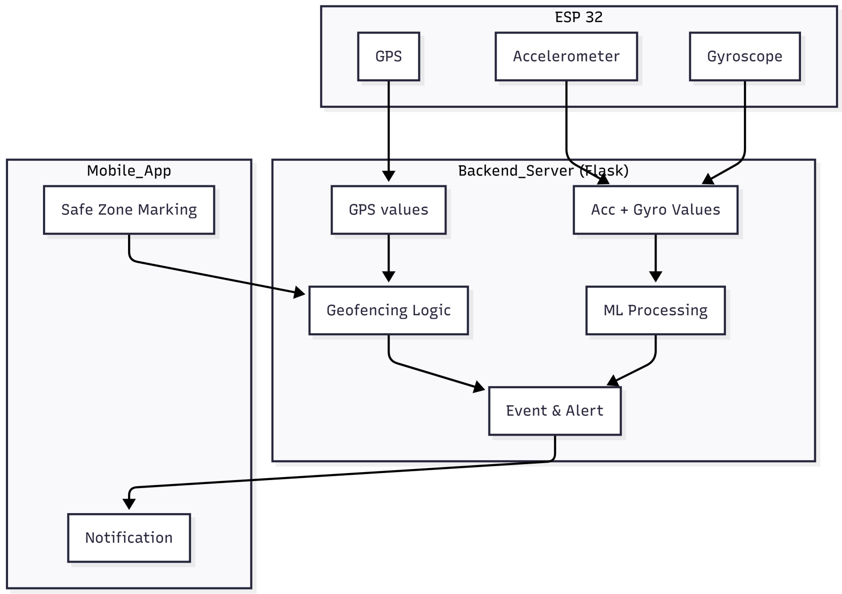


Figure 3.1: Architecture Diagram

3.2.2 Sequence Diagram

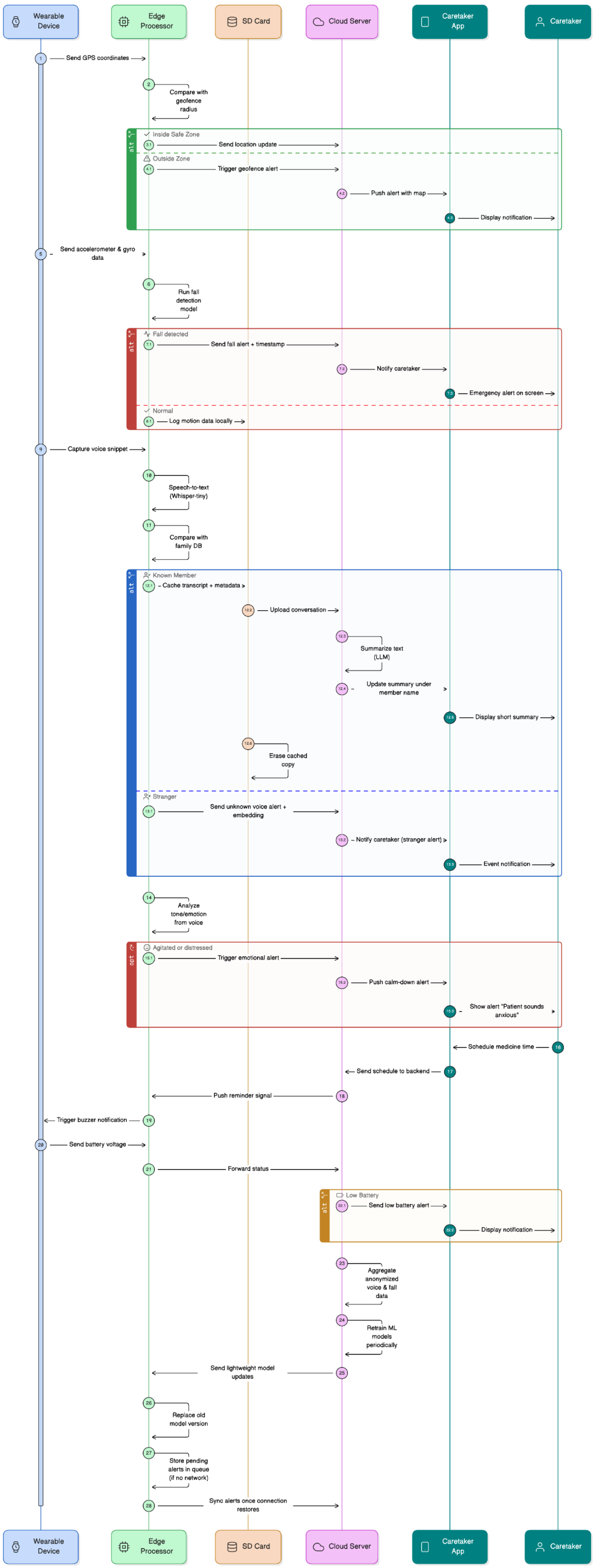
The sequence diagram illustrates the chronological flow of interactions between the main components of the Koode system during normal operation and emergency situations. It describes how data generated by the wearable device moves through the system and results in notifications being delivered to caregivers.

The process begins when the wearable device continuously collects motion data using embedded sensors such as the accelerometer and gyroscope. The microcontroller processes the sensor readings to monitor movement patterns and detect abnormal conditions such as sudden falls.

When a potential fall or emergency event is detected, the device prepares the relevant data and transmits it to the backend server through the communication module. The backend server receives the data and performs additional processing, such as storing the event in the database and verifying the alert condition.

Once the event is confirmed, the backend system triggers the notification service. Alerts are then sent to the registered caregivers through the mobile application using real-time messaging services. The mobile application displays the alert information, allowing caregivers to respond quickly.

The sequence diagram therefore demonstrates the step-by-step interaction between the wearable device, backend server, database, and mobile application, ensuring efficient detection, processing, and communication of emergency events within the Koode system.



Wearable Device
Edge Processor
SD Card
Cloud Server
Caretaker App
Caretaker
Send GPS coordinates
Compare with geofence radius
alt
Inside Safe Zone
Send location update
Outside Zone
Trigger geofence alert
Push alert with map
Display notification
Send accelerometer & gyro data
Run fall detection model
alt
Fall detected
Send fall alert + timestamp
Notify caretaker
Emergency alert on screen
Normal
Log motion data locally
Capture voice snippet
Speech-to-text (Whisper-tiny)
Compare with family DB
alt
Known Member
Cache transcript + metadata
Upload conversation
Summarize text (LLM)
Update summary under member name
Display short summary
Erase cached copy
Stranger
Send unknown voice alert + embedding
Notify caretaker (stranger alert)
Event notification
Analyze tone/emotion from voice
opt
Agitated or distressed
Trigger emotional alert
Push calm-down alert
Show alert "Patient sounds anxious"
Schedule medicine time
Send schedule to backend
Push reminder signal
Trigger buzzer notification
Send battery voltage
Forward status
alt
Low Battery
Send low battery alert
Display notification
Aggregate anonymized voice & fall data
Retrain ML models periodically
Send lightweight model updates
Replace old model version
Store pending alerts in queue (if no network)
Sync alerts once connection restores
Wearable Device
Edge Processor
SD Card
Cloud Server
Caretaker App
Caretaker

3.3 Module Division

3.3.1 Location Tracking and Geofencing Module

The Location Tracking and Geofencing Module provides ongoing spatial awareness of the patient's position using GPS coordinates captured by the wearable device. The module allows predefined safe zones ("geofences") within which the patient may safely move. These boundaries are set by caregivers via the mobile application or web portal and uploaded securely to the device.

The wearable device transmits location data at regular intervals to the cloud backend, where each update is evaluated against the defined safe zones. When the patient's location exceeds a boundary or enters a restricted area, caregivers instantly receive alerts through push notifications, SMS, or phone calls. This capability greatly mitigates risks related to wandering—a common and dangerous behavior in Alzheimer's patients—by facilitating rapid intervention.

The geofencing module supports dynamic zone configuration, multiple safe areas, and real-time alerts. The recorded location trails are stored securely for later review, audit, and analytics, helping caregivers and medical professionals assess risk and plan targeted interventions to ensure patient safety.

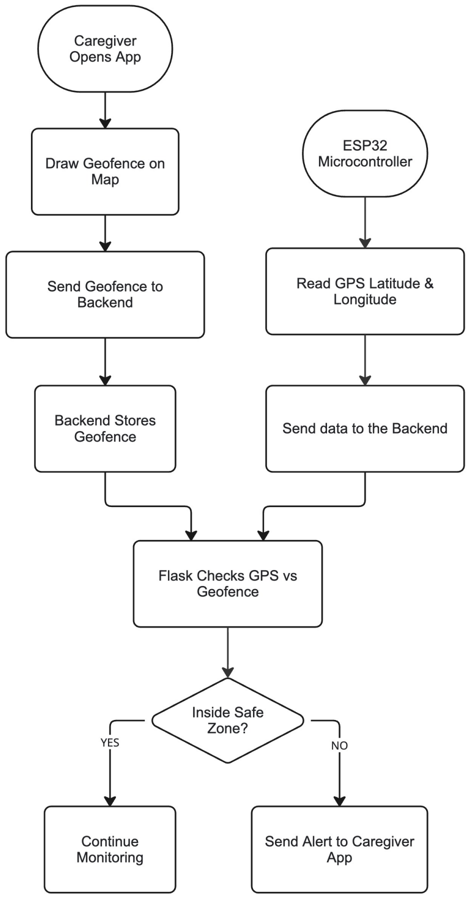


Figure 3.3: GPS and Geofencing

3.3.2 Fall Detection Module

The Fall Detection Module is engineered to reliably monitor and analyze motion data to identify actual fall events, distinguishing them from routine daily activities. This component utilizes a combination of accelerometer and gyroscope sensors embedded in the wearable device to capture three-dimensional linear and angular velocity signals generated from the patient's movements.

The sensor data streams are processed in real time by embedded algorithms, which may use either threshold-based or machine learning techniques for classification. Threshold-based algorithms monitor abrupt changes in acceleration magnitude and orientation to detect abnormal patterns characteristic of falls, such as a sudden peak followed by inactivity and a lying posture.

Machine learning models, including Support Vector Machines, Decision Trees, or Neural Networks, are trained using comprehensive datasets of sensor signals labeled as "fall" or “non-fall” to improve detection accuracy and reduce false alarms.

Upon detection of a potential fall, the system immediately triggers an audible alert on the device and sends notifications to caregivers via the cloud backend and mobile application, facilitating rapid assistance and improving patient safety.

The module emphasizes low-power and real-time operation, ensuring continuous monitoring without compromising battery life or device performance. Integration with GPS also assists in recording the patient’s location at the exact moment of the incident, providing caregivers with actionable information for emergency response.

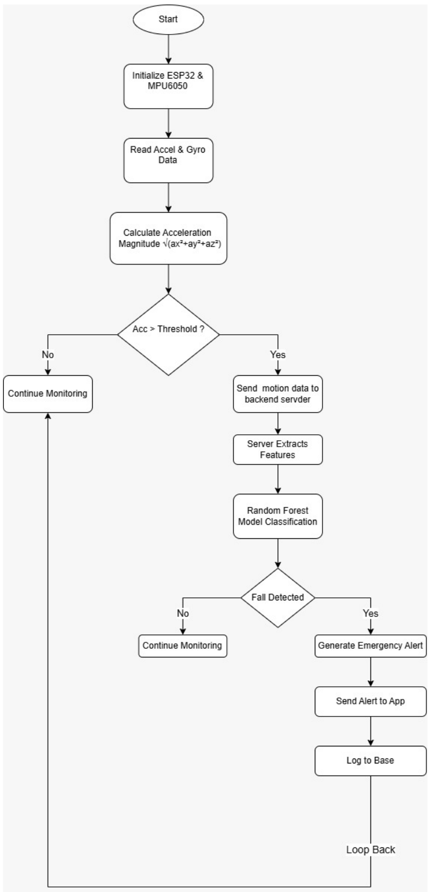


Figure 3.4: Fall Detection

3.3.3 Voice Recognition Module

The Voice Recognition Module transforms the wearable device into a conversational assistant and security system. It enables voice-based interactions for routine communication, reminders, emergency requests, and daily prompts, thereby supporting patients’ independence and engagement.

Technically, the module uses microphone input to capture speech, which is analyzed using on-device voice biometrics. The detected voice is compared against a stored database of known caregivers and family members using advanced voice embedding algorithms.

If an unknown voice is detected interacting with the patient, the system generates an alert to notify caregivers of possible unauthorized access or interaction. Additionally, caregivers can configure customized prompts, reminders, and messages that are delivered through the wearable device's speaker.

The voice interface is designed for ease of use, supporting both direct controls and context-sensitive commands to ensure accessibility for elderly users.

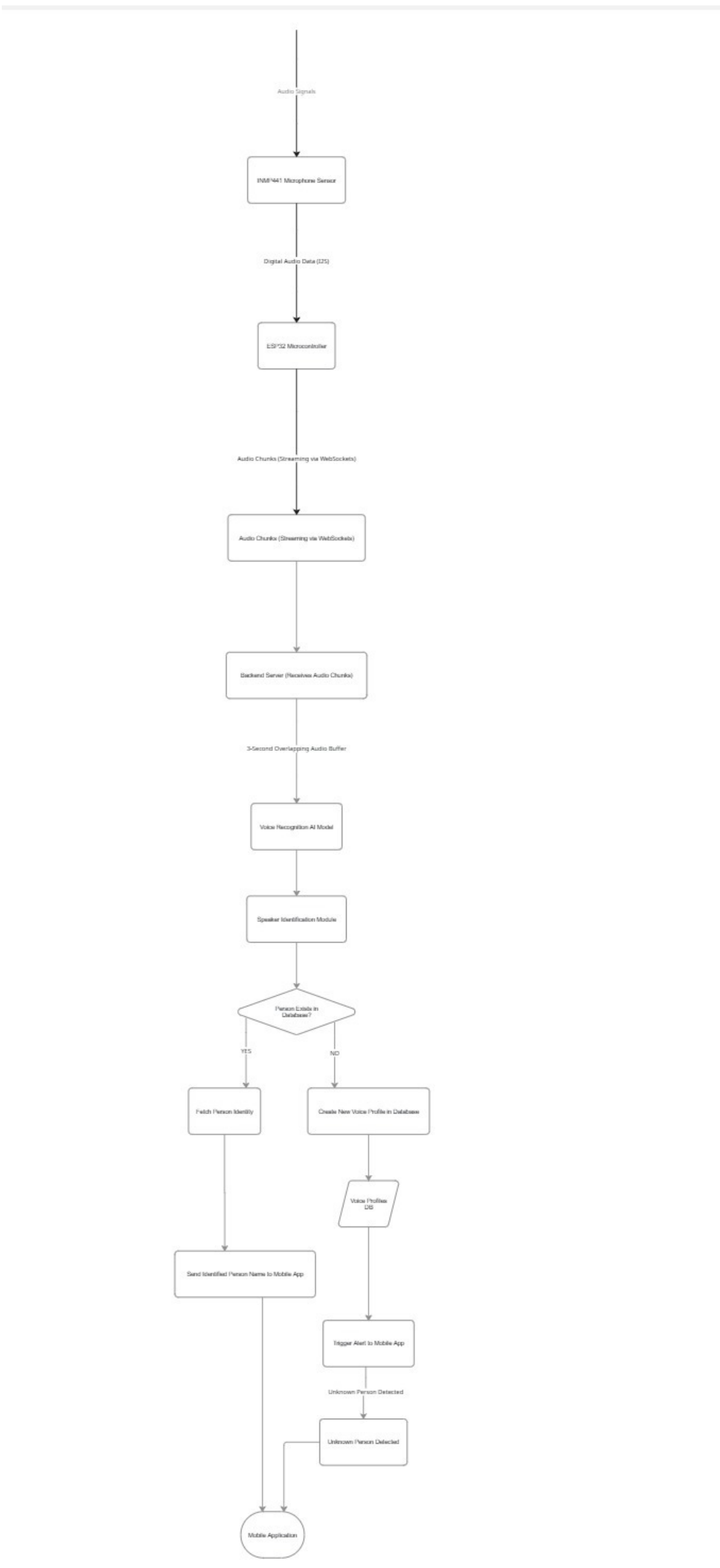


Figure 3.5: Voice Recognition Flowchart

3.3.4 IoT Integration

The IoT Integration Module acts as the backbone for seamless data transmission and synchronization across all system components. The wearable device—equipped with GPS, BLE, accelerometer, and microphone sensors—collects real-time physiological and environmental data from the patient.

This data is securely transmitted to the mobile or web application through BLE or Wi-Fi connectivity, allowing instant feedback, alerts, and scheduling functionalities for both patients and caregivers.

Data synchronization is further managed by the backend server using REST API or MQTT communication protocols. The server processes the incoming data for secure storage in a database, AI-powered analytics, and automated event or alert generation.

All processed information and alerts are made available on an open-source IoT cloud platform for advanced visualization and device management.

This module ensures reliable, low-latency connectivity and robust communication pathways between hardware devices, backend infrastructure, and cloud services. The architecture includes end-to-end data flow, encryption mechanisms, scalable cloud dashboards, and device control points that collectively enable comprehensive monitoring and proactive interventions for Alzheimer's patient safety.

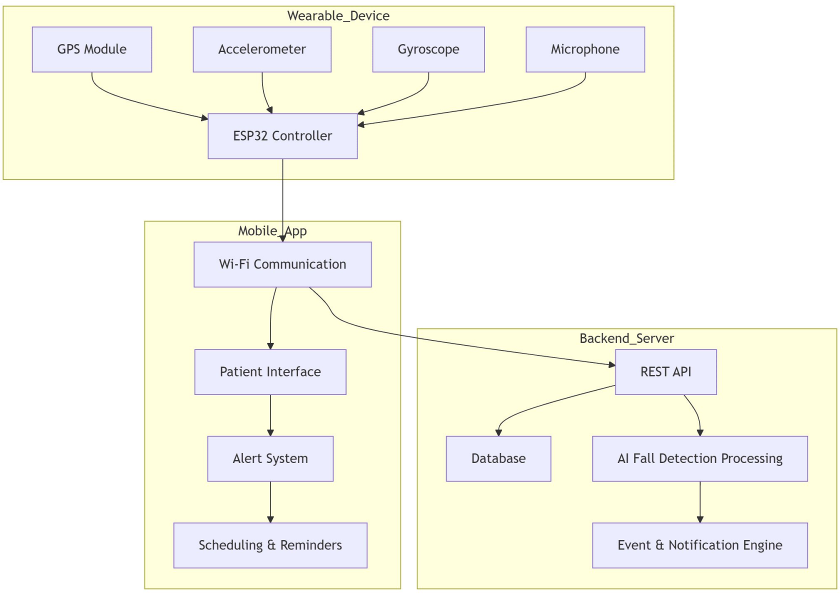


Figure 3.6: IoT Integration

3.3.5 Caregiver Mobile Application

The Caregiver Mobile Application serves as the central control and visualization hub for patient monitoring and system management. Developed for both Android and iOS platforms using frameworks such as Flutter or React Native, the application enables caregivers to effectively manage and monitor patients.

Key functionalities of the application include:

- Viewing the patient's live location on an interactive map.
- Receiving, managing, and acknowledging alerts such as fall detection, geofence breaches, and emergency button activations.
- Setting schedules, reminders, and personalized care routines tailored to the patient's daily needs.
- Accessing historical records including movement logs, alert events, and voice interaction history.
- Configuring device settings, defining safe zones, and managing authorized contacts remotely.

Through secure cloud integration, the application ensures that alerts and status updates are delivered promptly and reliably. It also allows multiple caregivers to monitor the same patient simultaneously and supports emergency escalation workflows to ensure timely response when critical situations occur.

3.4 Tools and Technologies

The Koode system integrates a combination of hardware components, software frameworks, communication technologies, and cloud-based services to ensure reliable monitoring and assistance for Alzheimer's patients. The following tools and technologies were used in the development of the system.

3.4.1 Embedded Hardware Platform

The wearable device is built using a microcontroller-based embedded system capable of collecting and transmitting sensor data. The platform integrates multiple sensors including a GPS module for location tracking, an accelerometer and gyroscope for motion monitoring, and a microphone for voice interaction.

These components enable real-time monitoring of patient activity and environmental conditions. The embedded hardware is designed for low power consumption to support long-duration operation in a wearable form factor.

3.4.2 Mobile Application Development

The caregiver mobile application is developed using cross-platform frameworks such as Flutter. These frameworks allow the application to run on Android devices using a single codebase.

The application provides an intuitive user interface for caregivers to monitor patient status, configure safe zones, receive alerts, and review historical activity data. Integration with cloud services ensures real-time updates and reliable notification delivery.

3.4.3 Backend and Cloud Infrastructure

A cloud-based backend infrastructure is used to manage data processing, storage, and communication between the wearable device and the caregiver application. The backend server handles incoming sensor data, performs data validation and processing, and generates alerts when abnormal conditions are detected.

Cloud services also store historical data such as location logs, fall detection events, and interaction records. This centralized architecture enables scalable monitoring and secure access to patient information.

3.4.4 Communication Protocols

Reliable communication between system components is achieved through wireless technologies with Wi-Fi. Wi-Fi enables internet connectivity for cloud synchronization.

Data transmission between the device, mobile application, and cloud backend is managed using REST APIs, ensuring efficient and low-latency communication.

3.4.5 Machine Learning Technologies

Machine learning algorithms are incorporated into the system to improve fall detection are used to classify motion data collected from accelerometer and gyroscope sensors.

These models are trained using labeled datasets containing examples of fall and non-fall events, allowing the system to distinguish between normal daily activities and dangerous incidents.

3.4.6 Development Tools

Several development tools were used during system implementation, including integrated development environments (IDEs), version control systems, and testing frameworks. Tools such as Arduino IDE were used for embedded programming, while modern code editors were used for mobile and backend development.

Version control systems such as Git were used to manage the project codebase and support collaborative development.

3.5 Work Breakdown and Responsibilities

The development of the Koode system was carried out collaboratively by the project team, with each member responsible for designing and implementing specific modules of the system. The work was divided based on the major functional components of the system to ensure efficient development and clear responsibility distribution.

3.5.1 Fall Detection Module

Anitta Seby was responsible for the design and implementation of the Fall Detection Module. This module focuses on detecting fall events using motion data collected from accelerometer and gyroscope sensors embedded in the wearable device. The work included sensor data acquisition, development of fall detection algorithms, and integration with the alert system to notify caregivers in case of emergencies.

3.5.2 Geofencing and Location Tracking

Angeline Tessa Saju handled the development of the Geofencing and Location Tracking Module. This component uses GPS technology to continuously monitor the patient's location and determine whether the patient remains within predefined safe zones. The responsibilities included implementing geofence boundary detection, location data processing, and alert generation when a boundary is crossed.

3.5.3 Voice Recognition Module

Ayush Alex Thomas was responsible for the Voice Recognition Module. This module enables voice-based interaction between the patient and the wearable device while also providing a security layer through voice identification. The work involved implementing voice input processing, voice recognition algorithms, and generating alerts when unknown voices are detected.

3.5.4 IoT Integration and System Connectivity

Christopher Cleatus was responsible for the IoT Integration Module. This module manages communication between the wearable device, cloud infrastructure, and the caregiver mobile application. Responsibilities included implementing wireless communication protocols, ensuring reliable data transmission, integrating cloud services, and enabling real-time monitoring and alert delivery.

3.6 Key Deliverables

The Koode project aims to develop a comprehensive wearable-based monitoring system for Alzheimer's patients. The project delivers a combination of hardware components, software applications, and integrated monitoring capabilities to ensure patient safety and caregiver assistance. The major deliverables of the project are described below.

3.6.1 Wearable Monitoring Device

The primary deliverable of the project is a wearable device designed for continuous monitoring of Alzheimer's patients. The device integrates multiple sensors including GPS for location tracking, accelerometer and gyroscope sensors for motion detection, and a microphone for voice interaction. The wearable device is designed to be lightweight, energy-efficient, and suitable for long-term usage by elderly patients.

3.6.2 Fall Detection System

A fall detection system is implemented within the wearable device to identify sudden falls or abnormal movements. Using motion sensor data and detection algorithms, the system analyzes acceleration and orientation changes to identify possible fall events. When a fall is detected, alerts are immediately generated and sent to caregivers to ensure timely assistance.

3.6.3 Geofencing and Location Tracking System

The project delivers a geofencing system that allows caregivers to define safe movement zones for the patient. The wearable device continuously monitors the patient's GPS location and compares it with predefined boundaries. If the patient exits the designated safe zone, the system automatically sends alerts to caregivers to enable rapid intervention.

3.6.4 Voice Recognition and Interaction Module

A voice recognition module is implemented to enable voice-based interaction with the wearable device. The system can recognize voices of authorized caregivers and family members while detecting unknown voices. If an unfamiliar voice is detected, the system generates a notification to alert caregivers of possible unauthorized interaction.

3.6.5 IoT-based Data Communication System

The project provides an IoT-enabled communication framework that connects the wearable device, cloud server, and caregiver mobile application. Sensor data collected by the wearable device is transmitted through wireless communication protocols and processed in the backend system. This infrastructure ensures reliable data transfer, real-time monitoring, and centralized storage of patient data.

3.6.6 Caregiver Mobile Application

A mobile application is developed to allow caregivers to monitor patient activity and manage system settings. The application provides features such as live location tracking, alert notifications, safe zone configuration, and access to historical data. The mobile interface is designed to be user-friendly and accessible for caregivers.

3.7 Project Timeline

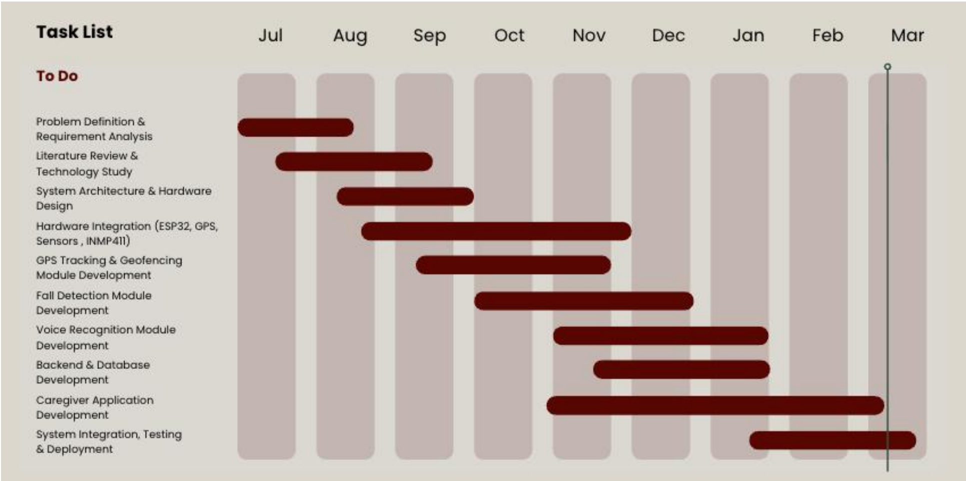


Figure 3.7: Gantt Chart

3.8 Summary of Chapter

This chapter presented the project planning and organizational aspects of the Koode system. It discussed the tools and technologies used for the development of the wearable monitoring system, including hardware components, software frameworks, and IoT communication technologies. The chapter also outlined the distribution of responsibilities among the team members, highlighting the specific modules handled by each contributor such as fall detection, geofencing, voice recognition, and IoT integration.

In addition, the key deliverables of the project were described, including the wearable device, monitoring modules, caregiver mobile application, and cloud-based backend system. Finally, the project timeline was introduced through a Gantt chart to illustrate the planned schedule of development activities. These elements collectively provide a structured overview of the resources, responsibilities, and workflow adopted for the successful execution of the project.

Chapter 4

Results and Discussions

This chapter presents the results obtained from the implementation of the Koode system and discusses the performance and effectiveness of the proposed solution. The evaluation focuses on the system's ability to monitor patient activity, detect critical events such as falls, and provide real-time alerts to caregivers. The results demonstrate how the integration of wearable sensing, cloud services, and mobile applications contributes to improved patient monitoring and safety.

4.1 System Implementation Results

The Koode system was successfully implemented using a wearable device integrated with motion sensors, GPS tracking, and communication modules. The wearable device continuously collected motion and location data and transmitted it to the backend system through the IoT communication infrastructure.

The backend server processed the received data, stored event logs in the database, and generated alerts whenever abnormal conditions such as fall detection or geofence violations occurred. The caregiver mobile application received these alerts in real time, allowing caregivers to monitor patient activity and respond quickly to emergencies.

Testing of the system confirmed that the wearable device could reliably capture motion data and detect sudden falls. Location tracking using GPS allowed caregivers to monitor the patient's position and receive notifications if the patient moved outside predefined safe zones.

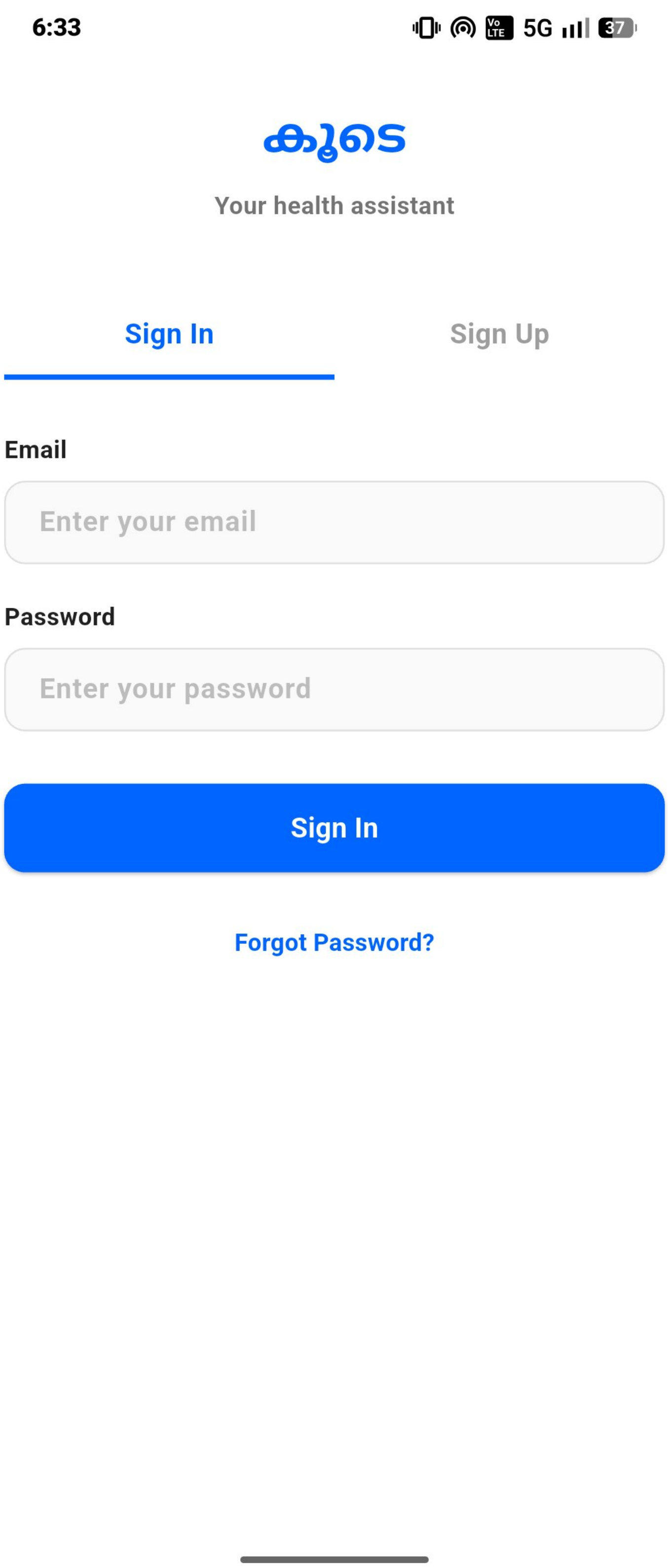


Figure 4.1: Login

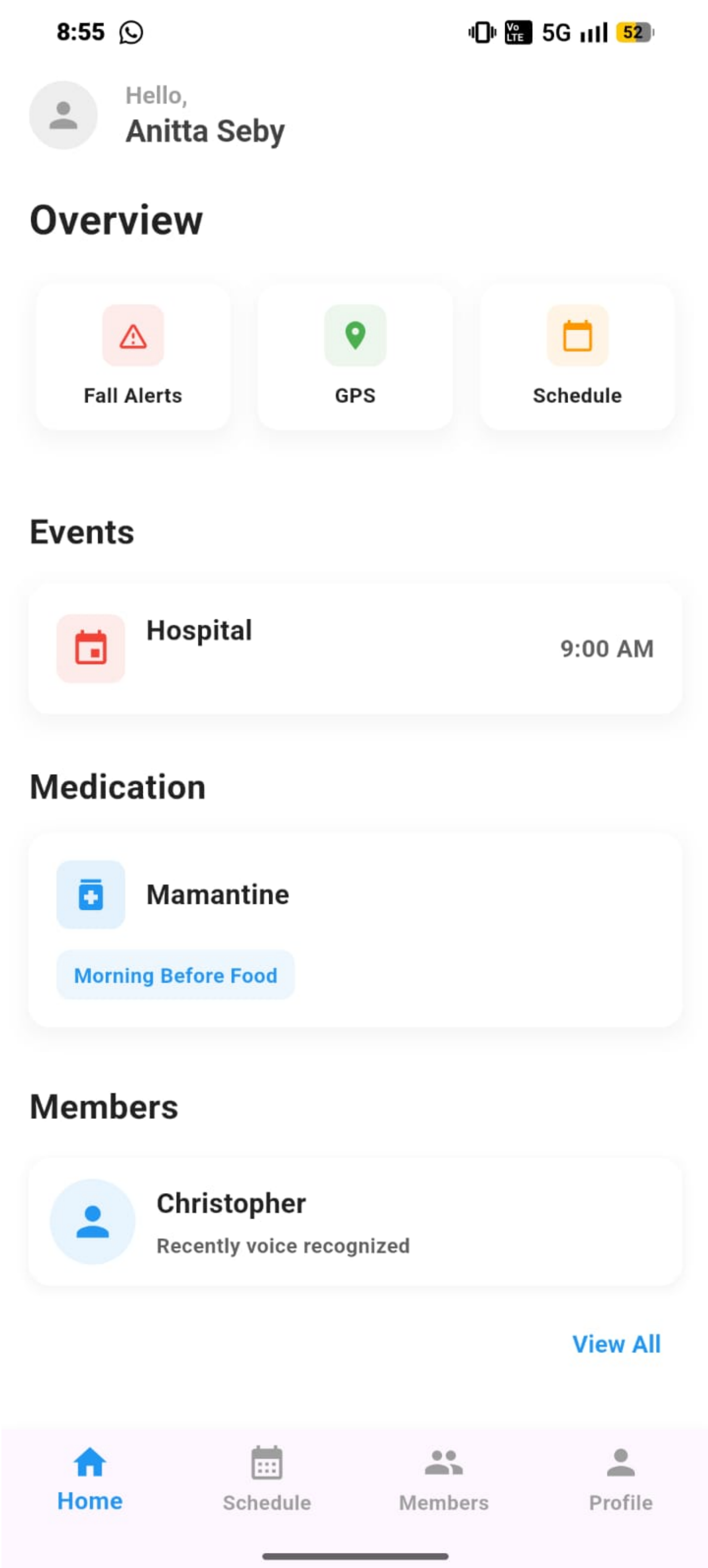


Figure 4.2: Dashboard

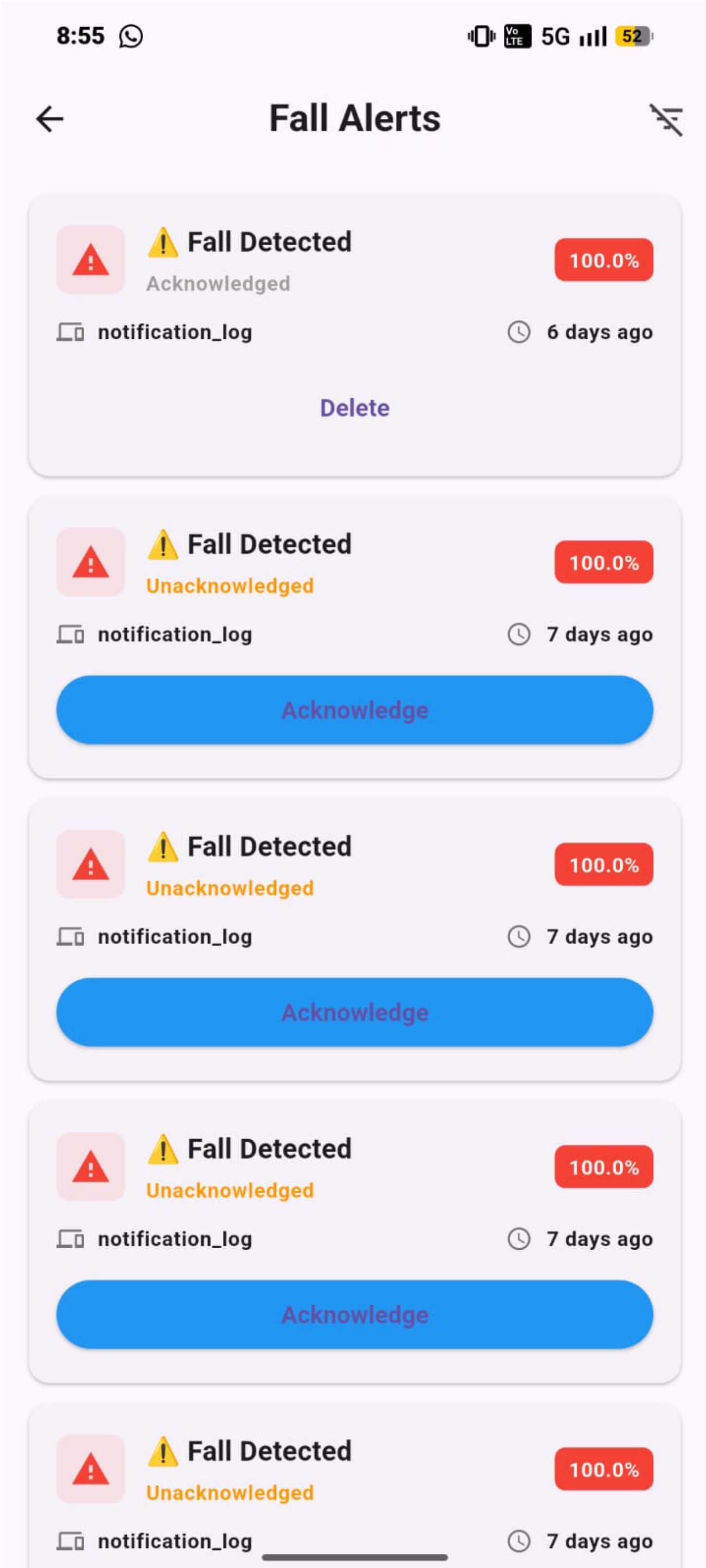


Figure 4.3: Fall alert logs

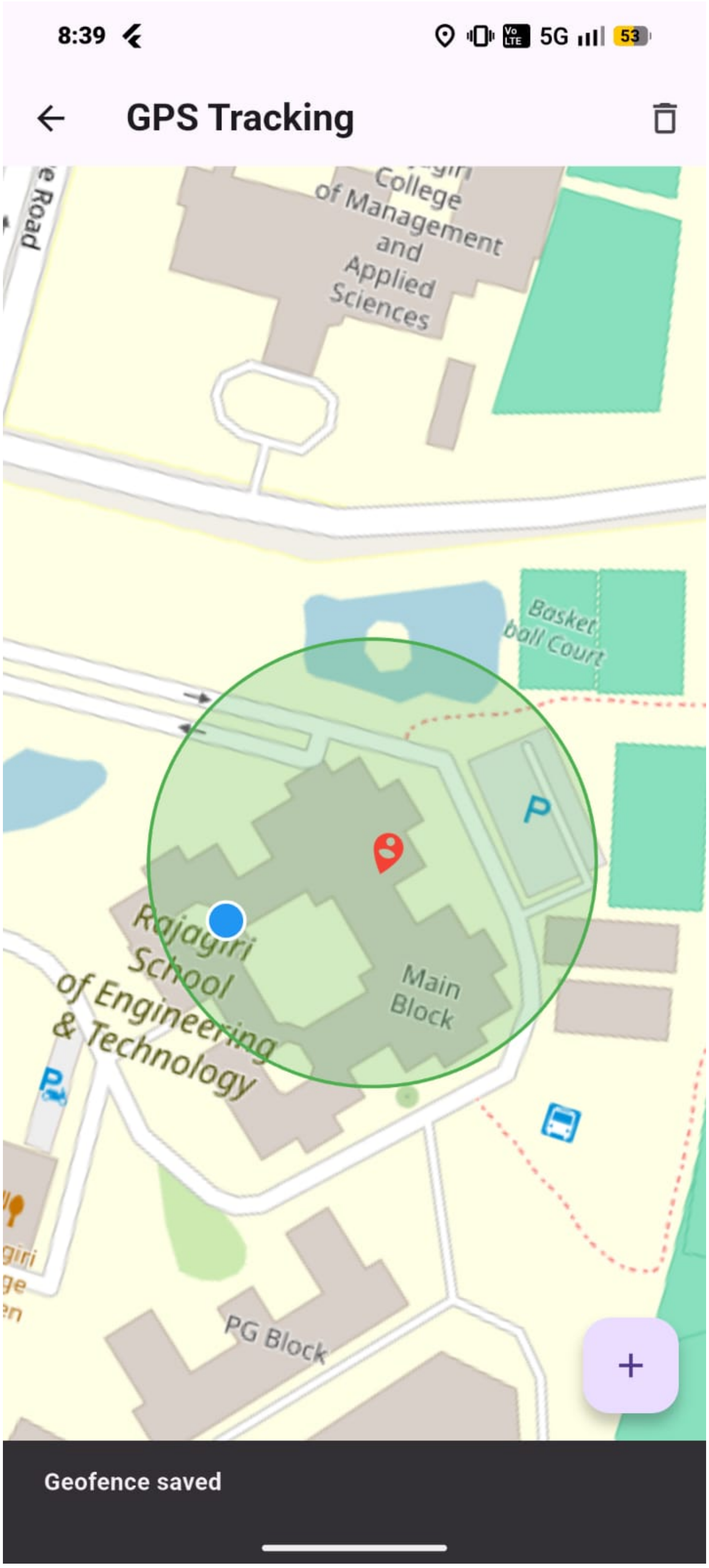


Figure 4.4: Setting Gefence Boundary

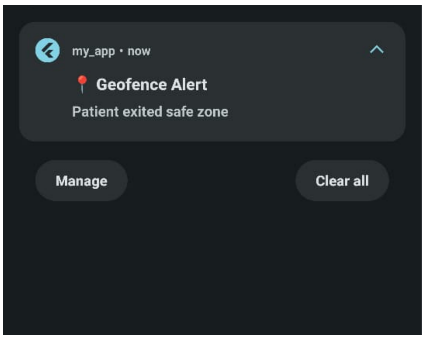


Figure 4.5: Geofence Notification Alert

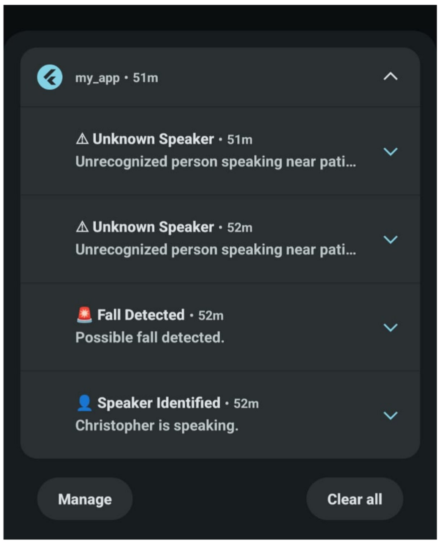


Figure 4.6: Notifications

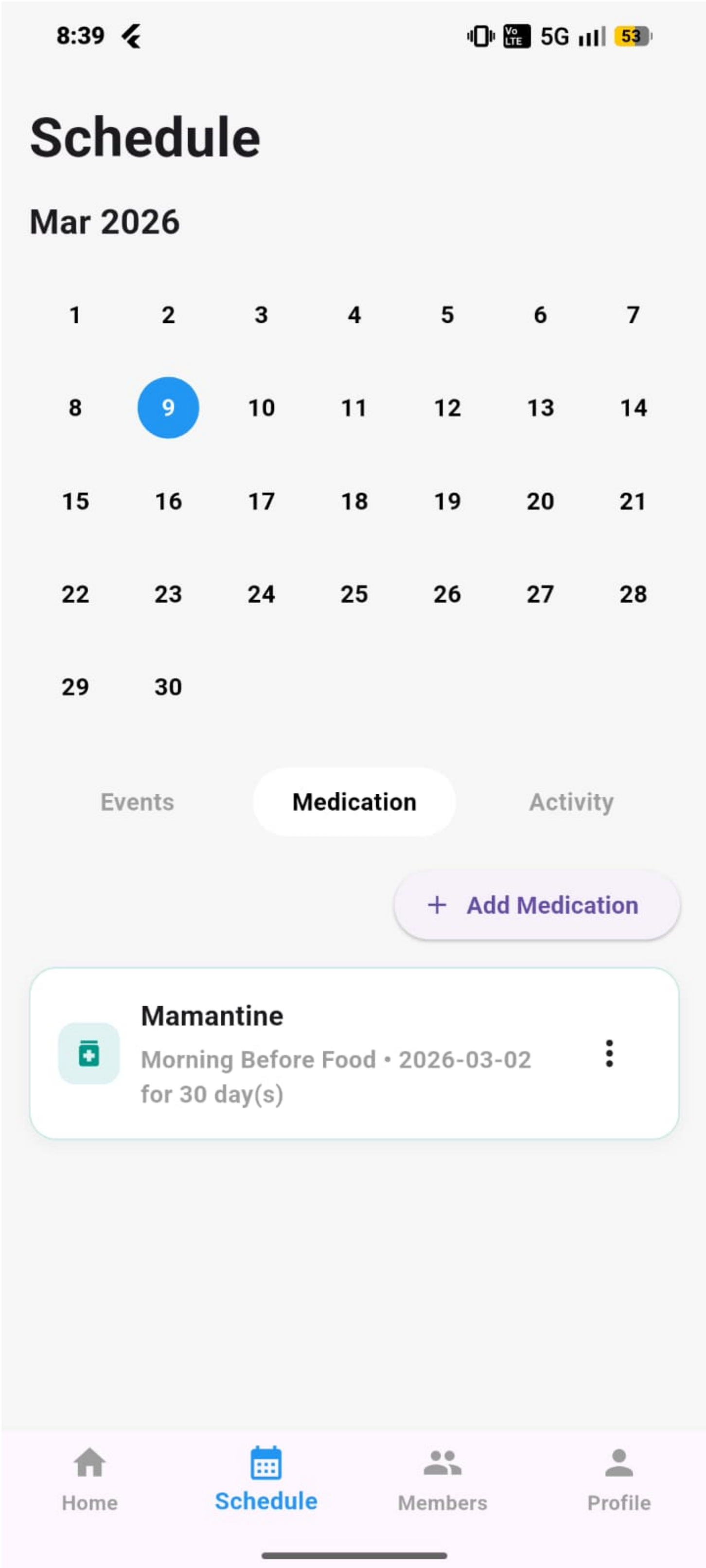


Figure 4.7: Medication Logs

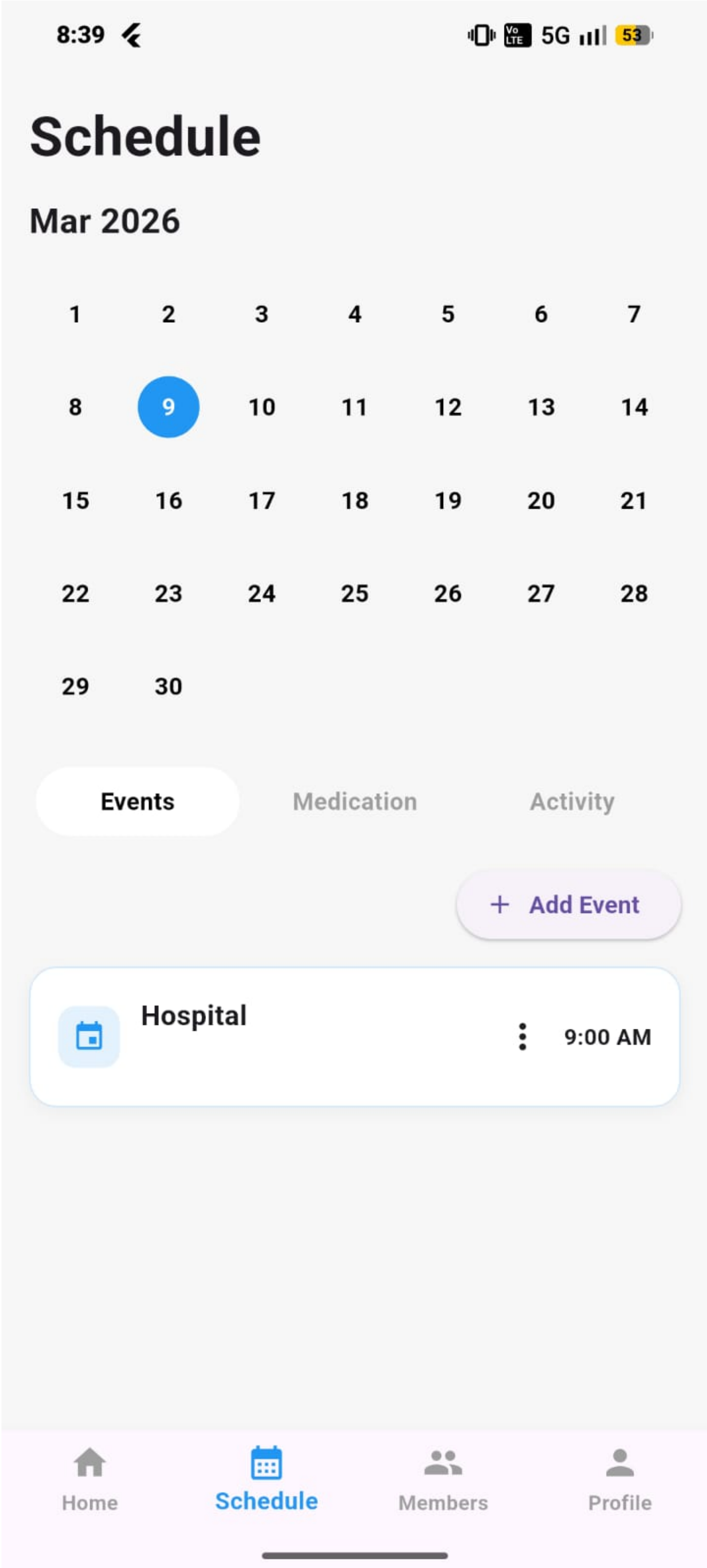


Figure 4.8: Event Logs

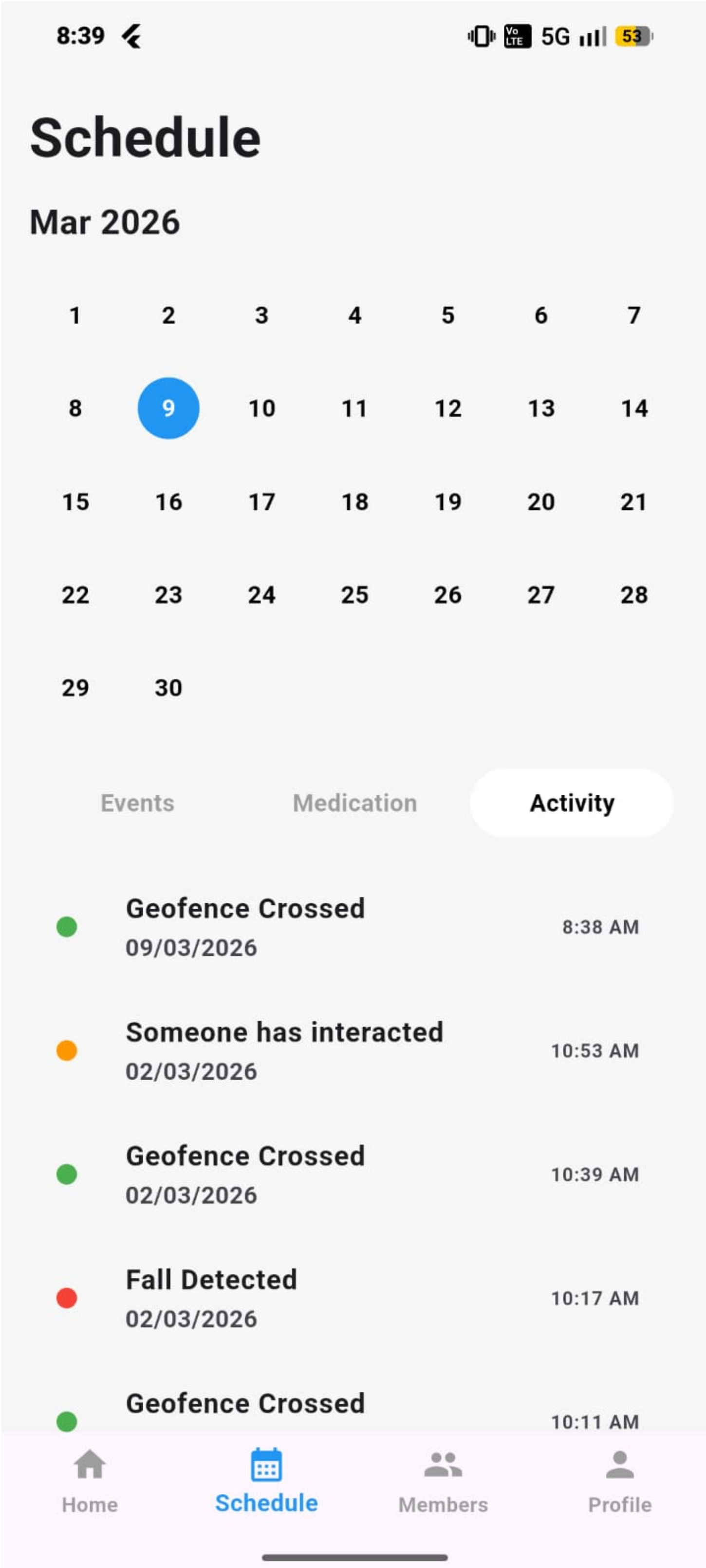


Figure 4.9: Activity Logs

4.2 System Performance and Analysis

The performance of the Koode system was evaluated based on responsiveness, reliability, and usability. The system demonstrated efficient real-time data transmission between the wearable device, cloud backend, and mobile application.

The fall detection module successfully identified abnormal motion patterns and generated alerts with minimal delay. Similarly, the geofencing system accurately detected boundary violations and notified caregivers promptly. The integration of cloud services ensured reliable data storage and continuous synchronization between devices.

The mobile application provided an intuitive interface for caregivers, enabling them to view live location updates, receive alerts, and access historical records of patient activity.

4.2.1 Discussion of System Effectiveness

The implementation results indicate that the Koode system can significantly improve patient monitoring and safety. By combining wearable sensing technology with cloud-based monitoring and mobile applications, the system provides continuous supervision of patients who may require assistance.

Real-time alerts enable caregivers to respond quickly to emergencies such as falls or wandering outside safe zones. The use of modular system architecture also allows the solution to be expanded in the future with additional sensors, improved analytics, or enhanced communication features.

Overall, the Koode system demonstrates the potential of IoT-based healthcare monitoring systems to support caregivers and enhance the safety and well-being of patients.

Chapter Summary

This chapter discussed the results obtained from the implementation of the Koode system and evaluated its performance in monitoring patient activity and generating alerts. The system successfully demonstrated real-time monitoring, fall detection, and geofencing capabilities through the integration of wearable devices, cloud services, and a caregiver mobile application. The results confirm the effectiveness of the proposed system in improving patient safety and caregiver response.

Chapter 5

Conclusions & Future Scope

5.1 Conclusion

This project presented the design and development of a smart wearable system intended to assist individuals in the early stages of Alzheimer's disease while supporting their caregivers. The system integrates multiple technologies such as voice recognition, fall detection using accelerometer and gyroscope sensors, GPS-based location tracking, and geo-fencing alerts. By combining embedded hardware, wireless communication, and a connected mobile application, the system is able to monitor patient activity and automatically notify caregivers when abnormal events occur.

The proposed solution aims to improve the safety and independence of patients while reducing the constant monitoring burden on caregivers. Real-time alerts, location tracking, and activity logs provide caregivers with continuous updates about the patient's condition. Overall, the project demonstrates how wearable technology and intelligent monitoring systems can play an important role in assisting people affected by memory-related disorders.

5.2 Future Scope

The system can be further improved and extended in several ways:

Integration of advanced machine learning algorithms to enhance the accuracy of fall detection and voice recognition.

Addition of health monitoring sensors such as heart rate, temperature, or oxygen level sensors for better patient health tracking.

Expansion of the system to support multiple patients and caregivers through a scalable cloud-based platform.

Integration with smart home systems to provide automated assistance and reminders for daily activities.

Improvement in hardware design to make the device more compact, energy-efficient, and comfortable for long-term use.

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Appendix A: Presentation

FINAL PRESENTATION

KOODE

A SMART COMPANION FOR ALZHEIMER'S PATIENTS

Guide:
Mr. Sebin Jose

Presented By:
Angeline Tessa Saju - U2203042
Anitta Seby - U2203045
Ayush Alex Thomas - U2203062
Christopher Cleatus - U2203071

Department of Computer Science and Engineering

2025-2026

Angeline Tessa Saju

Anitta Seby

Ayush Alex Thomas

Christopher Cleatus

The Team

Contents:

PROBLEM DEFINITION

Alzheimer's patients face:

- Memory loss and difficulty recognizing familiar people
- Disorientation and wandering outside safe zones
- High risk of falls and delayed emergency response
- Dependence on continuous caregiver supervision

Existing solutions are:

- Feature-specific (only tracking or only fall detection)
- Expensive or not integrated
- Lacking real-time intelligent assistance

There is a need for an integrated, wearable assistive system that ensures safety, monitoring, and cognitive support in one platform.

REAL-WORLD SCENARIO

Scenario 1:
An elderly patient steps outside home and forgets the way back.

Scenario 2:
The patient falls in a room with no one nearby.

Scenario 3:
A familiar person speaks, but the patient cannot recognize them.

OBJECTIVE

The objective of this project is to design and develop a secure, IoT wearable assistant that enhances the safety, independence, and communication of Alzheimer's patients.

- Design and develop an integrated wearable assistive system for Alzheimer's patients
- Detect falls using motion sensor data
- Monitor real-time location with geo-fencing alerts
- Identify familiar speakers using voice recognition
- Provide instant alerts to caregivers through a mobile application
- Enhance patient safety and reduce caregiver stress

Overall Goal:
To create an integrated and reliable assistive system that reduces caregiver burden while improving patient safety and quality of life.

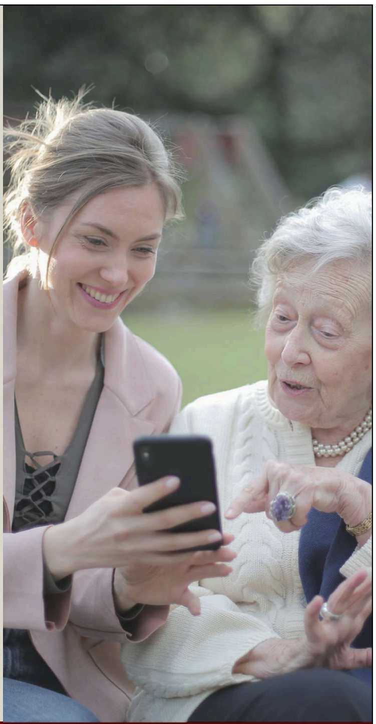
PURPOSE AND NEED:

Purpose

- To enhance the safety and independence of Alzheimer's patients.
- To provide real-time monitoring and emergency assistance.
- To support caregivers with timely alerts and information.
- To integrate multiple assistive technologies into one wearable system.

Need

- Rising number of Alzheimer's cases due to aging population.
- Increased risk of wandering and fall-related injuries.
- High emotional and physical burden on caregivers.
- Existing solutions are fragmented and not fully integrated.
- Demand for affordable and continuous monitoring systems.



LITERATURE SURVEY

Paper 1: Implementing GPS Trackers for People with Dementia at Risk of Wandering

- Objective: To test whether GPS trackers can reduce wandering risks in dementia patients.
- Participants: 45 dementia wearer–carer dyads.
- Method: Wearable GPS device provided to patients for 6 months; carers monitored movement via app.
- Key Features of Device:
  - Real-time tracking
  - Geofencing (alerts when outside safe zone)
  - Fall Alerts
  - Two-way communication (SOS button + speaker)

LITERATURE SURVEY

Paper 1: Implementing GPS Trackers for People with Dementia at Risk of Wandering

Patient wears GPS tracker

↓

Location sent to server

↓

Carer monitors via app

↓

Alerts triggered if geofence crossed

LITERATURE SURVEY

Paper 1: Implementing GPS Trackers for People with Dementia at Risk of Wandering

Advantages:

- Improved safety — no patients came to harm during wandering.
- Increased independence for wearers → more freedom to move around safely.
- Quick retrieval — carers could locate patients without calling police.
- Reduced carer burden.

Disadvantages / Limitations:

- Small sample size (45 dyads).
- Most patients were in moderate-to-severe dementia → couldn't self-report.
- No control group → hard to prove GPS alone caused the benefit.

LITERATURE SURVEY

Paper 2: CareFall – Automatic Fall Detection through Wearable Devices and AI Methods

- Wearable-Based Monitoring – Uses smartwatch sensors (3-axis accelerometer + gyroscope) for continuous fall detection.
- Dual Approaches – Compares threshold-based and machine-learning-based classification for recognizing falls vs. daily activities.
- Feature Extraction – Derives statistical features from sensor signals.
- Performance – Machine learning (SVM, RF, ANN) with combined accelerometer + gyroscope achieves accuracy up to 98.4%, outperforming threshold methods.
- Practical Impact – Provides user-friendly, automatic alerts to emergency services, aiming to replace button-based systems.

LITERATURE SURVEY

Paper 2: CareFall – Automatic Fall Detection through Wearable Devices and AI Methods

Advantages

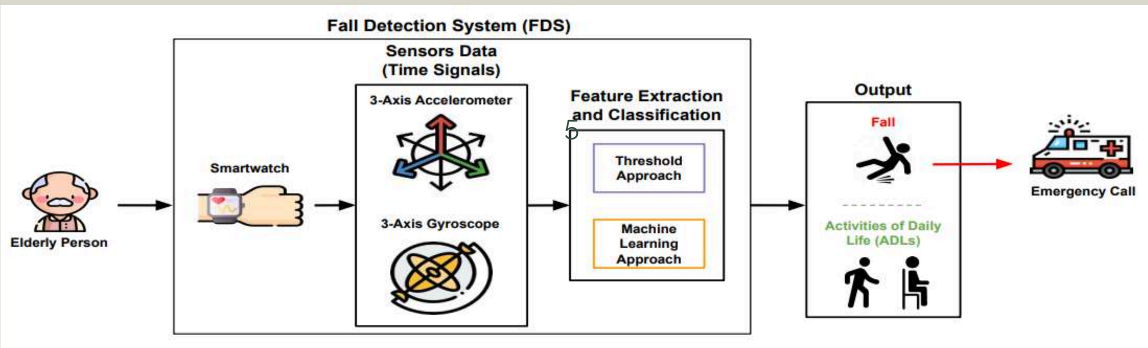
- Provides automatic fall detection without manual intervention .
- High accuracy when using ML with accelerometer + gyroscope features.
- Works on commercial smartwatches, making it affordable and user-friendly.
- Real-time detection enables faster emergency response.

Limitations

- Threshold methods prone to false alarms despite simplicity.
- Performance depends on sensor placement (wrist only tested).
- Battery and connectivity constraints of smartwatches may limit continuous monitoring.

LITERATURE SURVEY

Paper 2: CareFall – Automatic Fall Detection through Wearable Devices and AI Methods



LITERATURE SURVEY

Paper 3: Generalized End-to-End Loss for Speaker Verification

- Proposes the GE2E loss, improving both accuracy and training speed for speaker verification models by updating the network using all pairs within a batch.
- Each batch: N speakers, M utterances per speaker—model pushes each embedding closer to its speaker's centroid and away from others using a similarity matrix.
- Softmax and contrast losses tested; GE2E outperforms tuple-based approaches, leading to faster convergence and lower EER

LITERATURE SURVEY

Paper 3: Generalized End-to-End Loss for Speaker Verification

- **Advantages:**
  - Achieves >10% EER reduction compared to TE2E, with training time cut by 60%.
  - Eliminates the need for manual tuple selection during training, making the process scalable.
  - MultiReader technique enables domain adaptation by training on multiple datasets (e.g., different keywords and dialects) simultaneously.
  - Supports both text-dependent and text-independent speaker verification tasks efficiently.
- **Limitations:**
  - Requires batches containing multiple speakers and utterances—needs sufficiently large diverse datasets.
  - Training stability can be affected by batch composition; careful batch construction is necessary.

LITERATURE SURVEY

Paper 3: Generalized End-to-End Loss for Speaker Verification

- **Methodology:**
  - Features (log-mel-filterbank energies) extracted from audio; embeddings generated using 3-layer LSTM, followed by projection and normalization.
  - For training: GE2E loss computed on a similarity matrix comparing each embedding to all centroids.
  - For inference: d-vectors from sliding windows are averaged to create an utterance-level embedding; cosine similarity scores used for verification.
  - GE2E loss combines efficient learning, end-to-end neural embeddings, and robust handling of speaker verification datasets, setting a new standard for SV model training.

LITERATURE SURVEY

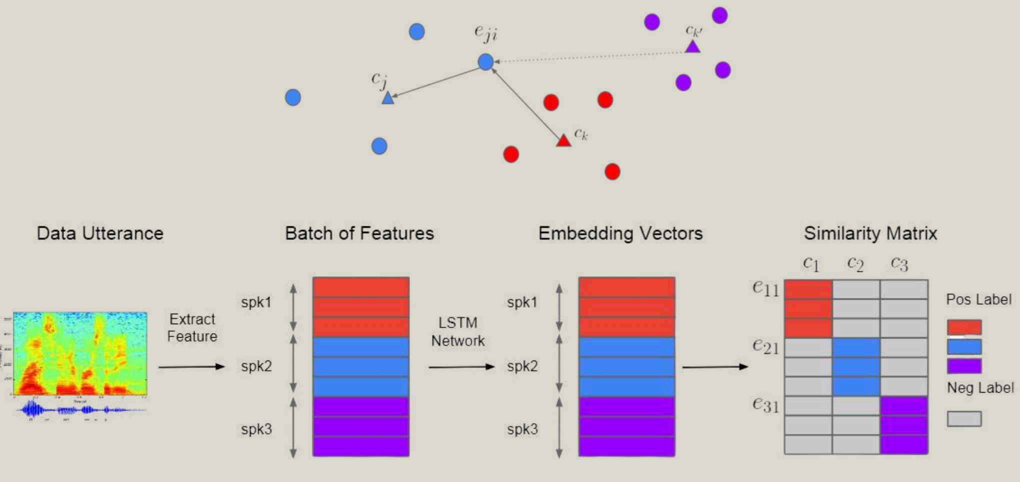


Fig. 1. System overview. Different colors indicate utterances/embeddings from different speakers.

LITERATURE SURVEY

Paper 4: Secure IoT Assistant-Based System for Alzheimers Disease

- Intelligent Support System – Uses IoT and AI (CNN + voice assistant) to provide Alzheimer's patients with safe, independent living and improved communication.
- Voice Interaction – Google Assistant integration helps patients interact naturally, reducing isolation and supporting mental well-being.
- CNN-based Recognition – Identifies family members vs. strangers with high accuracy, ensuring safety in social interactions.
- Real-time Safety Features – GPS tracking with geofencing and alerts, provide quick caregiver response.

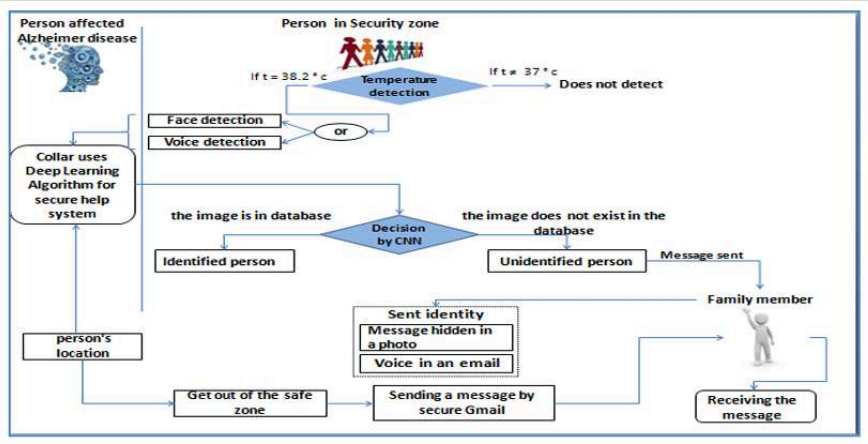
LITERATURE SURVEY

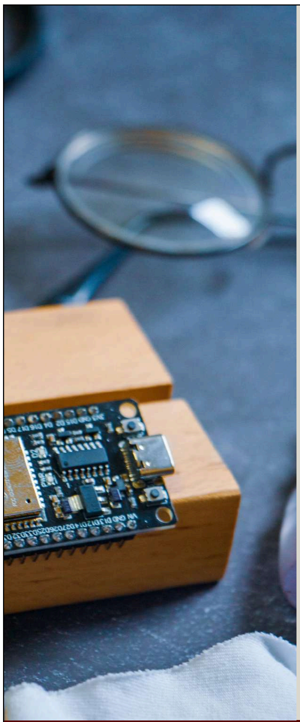
Paper 4: Secure IoT Assistant-Based System for Alzheimers Disease

- **Advantages**
  - Provides 24/7 monitoring and safety for Alzheimer's patients.
  - Voice interaction makes the system user-friendly and reduces social isolation.
  - Real-time GPS + geofencing ensures quick caregiver alerts in case of wandering.
  - Runs on low-cost IoT hardware with open-source tools, making it affordable.
- **Limitations**
  - Device dependency – patients may forget or refuse to wear the device.
  - False positives/negatives in fall detection or voice recognition can cause confusion.
  - Battery limitations restrict continuous monitoring.
  - Internet/GPS unavailability may affect performance outdoors or in remote areas.

LITERATURE SURVEY

Paper 4: Secure IoT Assistant-Based System for Alzheimers Disease





PROPOSED SYSTEM

Our proposed system is an IoT-based wearable assistant designed to provide:

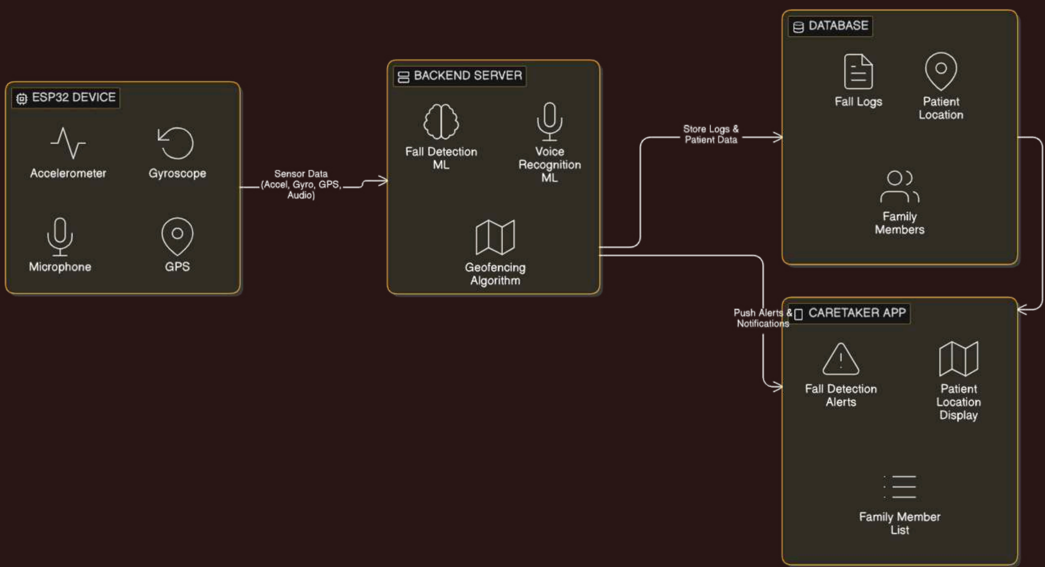
- Real-time fall detection using motion sensor data
- Continuous GPS tracking with geo-fencing alerts
- AI-based voice recognition for speaker identification
- Cloud-based data processing and storage
- Real-time caregiver monitoring through a mobile application

The system integrates hardware, AI, and mobile technologies into a unified assistive platform.

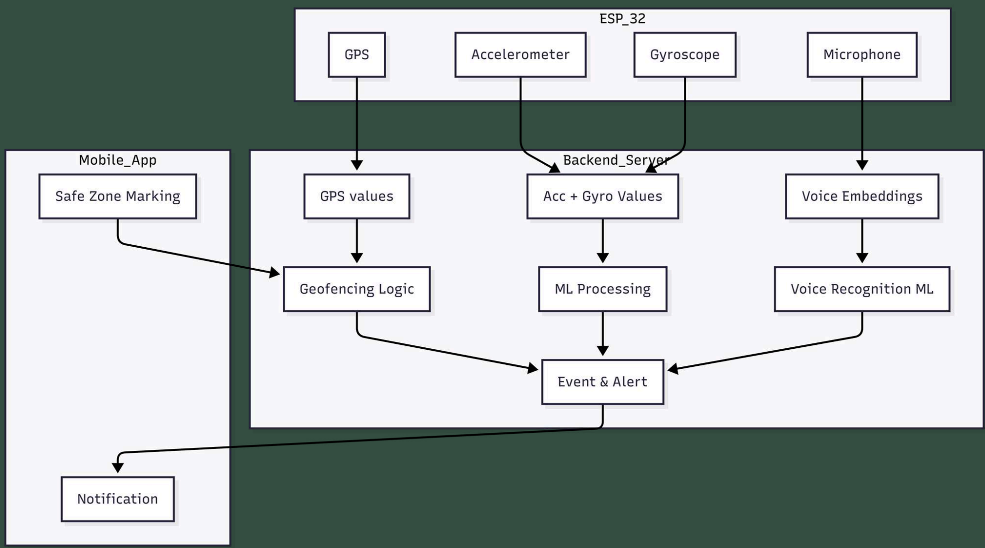
COST ESTIMATION

HARDWARE	COST
ESP32 Dev Module	₹450
MPU6050 (Accelerometer + Gyroscope sensor)	₹300
Neo 6M (GPS Module)	₹280
INMP441 (Microphone)	₹250
Miscellaneous (cables, breadboard)	₹200
TOTAL	₹2000

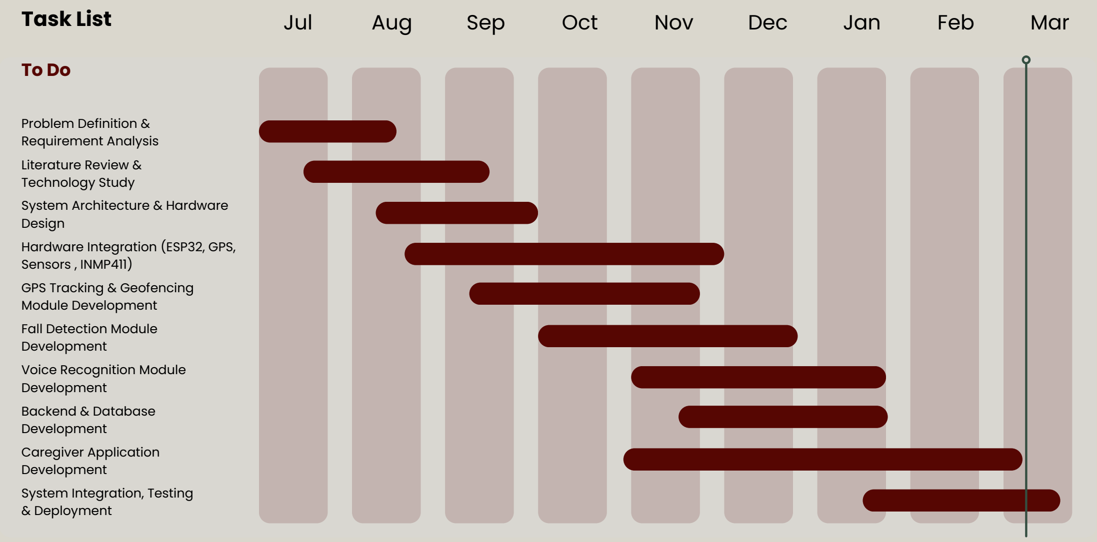
SYSTEM ARCHITECTURE



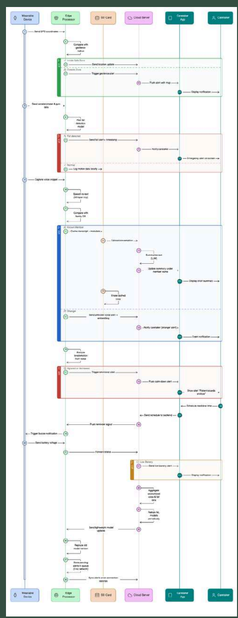
SYSTEM ARCHITECTURE



GANTT CHART



SEQUENCE DIAGRAM



MODULE DIVISION

The proposed IoT-based Smart Wristband System is divided into the following core modules:

- Location tracking and Geofencing. Module
- Fall Detection Module
- Voice Recoginition Module
- Caregiver Mobile Application Module

These modules work together to provide continuous monitoring, intelligent analysis, and real-time alert communication between the patient and caregiver.

1. Location Tracking & Geofencing Module

This module continuously captures GPS data from the wristband and monitors the patient's movement relative to predefined safe zones. If the patient crosses the boundary, an alert is triggered and sent to the caregiver application.

2.Fall Detection Module

This module collects accelerometer and gyroscope data from the wristband sensors and uses a machine learning model to analyze abnormal motion patterns. If a fall is detected, an emergency notification is generated and send to the caretaker app.

3.Voice Recognition Module

This module enables patient–assistant communication through voice input. It processes audio data for recognition and assists in interaction while maintaining conversation logs for caregiver monitoring.

4. Caregiver Mobile Application Module

This module provides real-time monitoring, alert notifications, location tracking, and system updates through a user-friendly mobile interface for caregivers.

ROLES ASSIGNED

Location Tracking and Geofencing Module

Angeline Tessa Saju

Fall Detection Module

Anitta Seby

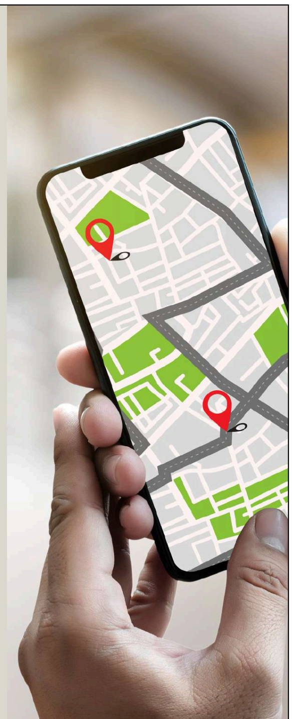
Voice Recognition Module

Ayush Alex Thomas

Mobile Application Devolepment

Christopher Cleatus

1.Location Tracking and Geofencing Module



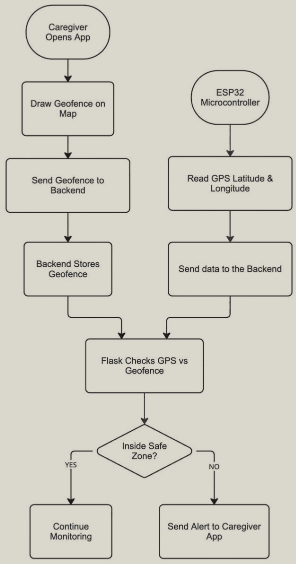
Location Tracking

- GPS module acquires real-time latitude and longitude coordinates
- Coordinates are transmitted to the backend
- The latest GPS location is maintained in the backend for real-time monitoring and geofence validation

Geo-Fencing Mechanism

- A predefined safe-zone radius is configured (e.g., 100 meters)
- The system calculates distance between current location and safe-zone center
- Distance is computed using the Haversine Formula
- If computed distance > defined radius → Geo-fence breach detected
- Immediate alert with map coordinates sent to caregiver application

1.Location Tracking and Geofencing Module



2.Fall Detection Module

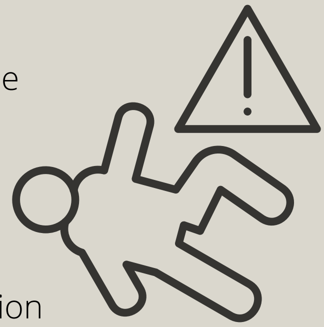
Hardware Implementation

- Used MPU6050 (3-axis accelerometer + 3-axis gyroscope)
- Connected sensor to ESP32 microcontroller
- Continuously captures real-time motion data
- Acceleration magnitude calculated to detect sudden impact events
- Abnormal motion data sent to cloud for ML-based classification
- Designed for low-power, compact wristband integration
- if fall detected, alert notification sent to caregiver mobile application

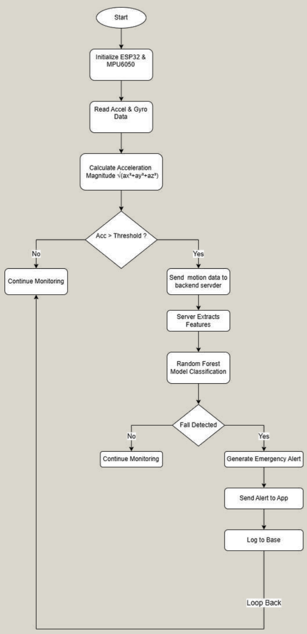
2.Fall Detection Module

Model Training Phase:

- Collected motion dataset containing fall and non-fall activities
- Preprocessed sensor data (noise filtering and normalization)
- Extracted statistical features from accelerometer and gyroscope readings
- Features included acceleration magnitude, peak values, mean, standard deviation, and orientation change
- Split dataset into 80% training and 20% testing
- Trained a Random Forest Classifier for fall vs non-fall classification
- Model learned movement patterns distinguishing falls from daily activities (walking, sitting, bending)
- Achieved reliable classification accuracy during testing



2.Fall Detection Module



FALL DETECTION DATASET

	A	B	C	D	E	F	G	H	I
1	SubjectID	ActivityID	Acc_X	Acc_Y	Acc_Z	Gyr_X	Gyr_Y	Gyr_Z	
2	1	1	-673.868	747.7292	-3952.06	21.57852	26.31638	32.37755	
3	1	1	-678.142	579.0758	-3961.02	13.47917	21.95423	21.76347	
4	1	1	-666.671	649.1164	-3975.28	33.97529	28.69301	35.69009	
5	1	1	-693.873	614.7042	-3961.87	18.18738	24.91807	24.55706	
6	1	1	-689.426	625.163	-3967.58	32.11036	25.93428	35.02834	
7	1	1	-701.661	590.4005	-3968.27	22.09149	25.3009	17.95658	
8	1	1	-693.767	642.27	-3976.84	24.39316	27.53086	30.36756	
9	1	1	-705.874	624.1895	-3975.45	6.027184	19.38143	23.5916	
10	1	1	-692.439	625.2115	-3976.9	11.33357	23.17128	32.33485	
11	1	1	-684.793	593.91	-3961.74	12.68242	19.16148	20.29336	
12	1	1	-678.734	637.3865	-3972.28	27.68052	20.59196	30.92587	
13	1	1	-683.1	614.5465	-3965	19.16783	17.76092	19.39352	
14	1	1	-681.708	639.1749	-3974.89	27.60942	23.38114	30.23941	
15	1	1	-697.34	618.5425	-3967.98	13.41548	19.20199	20.44275	
16	1	1	-692.62	641.571	-3967.02	17.60125	23.16162	34.73486	
17	1	1	-696.425	615.129	-3965.24	7.416491	14.75668	24.12196	
18	1	1	-681.796	641.3093	-3964.03	19.90771	21.55226	33.22227	
19	1	1	-685.963	623.5821	-3958.41	17.99166	22.54965	24.34244	
20	1	1	-673.071	648.6557	-3982.18	19.94351	22.29055	36.72563	
21	1	1	-680.004	599.5827	-3972.39	5.353314	18.96479	22.28254	

FallAllD_Wrist_AccGyr

	A	B	C	D	E	F	G	H	I
1048556	9	19	3912.225	67.49474	-1060.7	437.5045	-152.32	1250.036	
1048557	9	19	2917.697	86.647	-902.971	-128.306	-184.7	1214.755	
1048558	9	19	2380.297	-129.392	-870.658	-747.738	-237.101	1243.799	
1048559	9	19	2148.324	-446.655	-884.796	-1038.31	-220.391	1349.71	
1048560	9	19	2211.365	-646.39	-1024.76	-1065.64	-153.46	1519.07	
1048561	9	19	2425.498	-837.01	-1119.63	-1006.33	-94.7584	1746.065	
1048562	9	19	2840.133	-1093.4	-1256.2	-959.716	-36.0049	1960.839	
1048563	9	19	3183.911	-1255.45	-1278.13	-913.261	17.70465	2084.963	
1048564	9	19	3435.713	-1502.14	-1162.62	-830.689	44.21017	2044.841	
1048565	9	19	3846.665	-1793.84	-1046.63	-530.832	-7.83296	1883.958	
1048566	9	19	4812.054	-2058.18	-1240.08	-72.8408	24.51728	1578.361	
1048567	9	19	5076.102	-2195.51	-1188.78	177.5022	15.74539	1025.966	
1048568	9	19	5228.32	-2078.21	-1031.11	452.3231	-29.3	401.262	
1048569	9	19	5237.847	-1414.39	-895.505	773.9156	-156.779	-121.955	
1048570	9	19	7198.95	-816.328	-1099.91	794.9331	-222.745	-421.05	
1048571	9	19	7123.159	-914.112	-1537.8	91.48952	6.262209	-857.039	
1048572	9	19	5068.039	-1228.45	-1037.76	-398.181	9.159231	-1300.78	
1048573	9	19	5064.473	-1028.57	-639.222	-276.462	-69.5973	-1611.71	
1048574	9	19	5568.962	-776.925	-554.799	-46.9645	-6.40933	-1795.88	
1048575	9	19	5191.065	-789.399	-430.719	208.9283	72.42609	-1925.12	
1048576	9	19	4657.302	-810.613	-369.414	517.5999	91.9877	-2019.63	

FallAllD_Wrist_AccGyr

3.Voice Recognition Module

Model Training Phase:

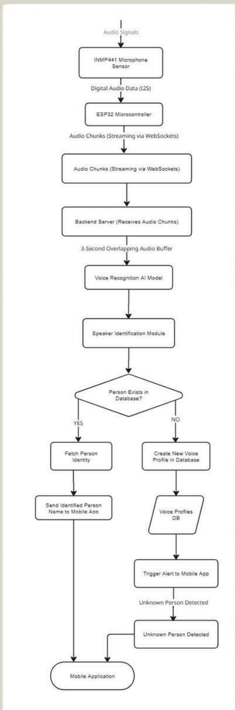
- Collected voice samples from multiple individuals
- Preprocessed audio data (noise reduction, normalization)
- Converted raw audio into feature representations (embeddings)
- Trained a supervised ML model for speaker classification
- Model learns unique voice patterns of each person
- Achieved reliable identification accuracy during testing

3.Voice Recognition Module

Hardware Implementation

- Used INMP441 digital MEMS microphone
- Connected to ESP32
- Captures high-quality digital audio input
- Optimized for compact wearable device integration
- Voice input captured through INMP441
- Audio segmented into 8-second windows
- Each segment converted into embeddings

3.Voice Recognition Module



4. Caregiver Mobile Application Module

Mobile Application Overview

- Developed using Flutter
- Provides real-time monitoring of patient status
- Communicates with backend using WebSockets
- Stores and retrieves data from Firebase database

Real-Time Communication (WebSockets)

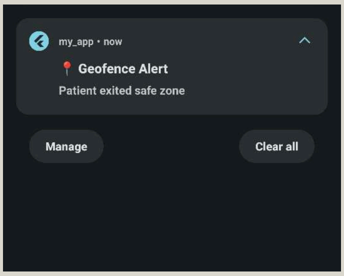
- WebSocket connection established between mobile app and backend server
- Enables continuous two-way communication
- Backend pushes real-time data:
- GPS updates
- Fall detection status
- Voice recognition results
- Eliminates repeated API polling
- Ensures low-latency alert delivery

4. Caregiver Mobile Application Module

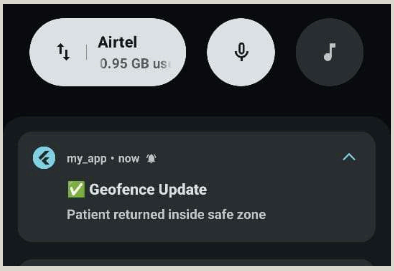
FCM (Firebase Cloud Messaging) Notification System

- On app registration, device generates unique FCM token
- FCM token stored in Supabase database
- When emergency event detected:
    - Backend retrieves caregiver's FCM token
    - Push notification sent via Firebase Cloud Messaging
- Alert delivered even if app is closed or in background

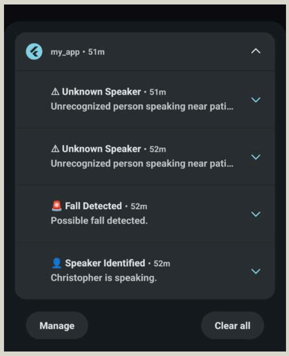
RESULTS AND OUTPUTS



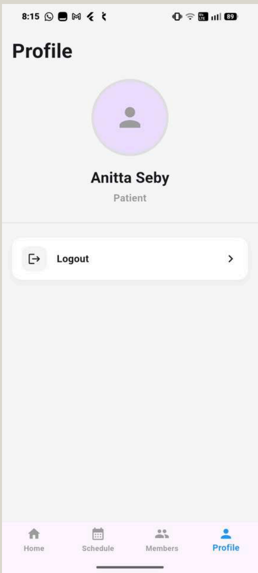
Patient exited Alert

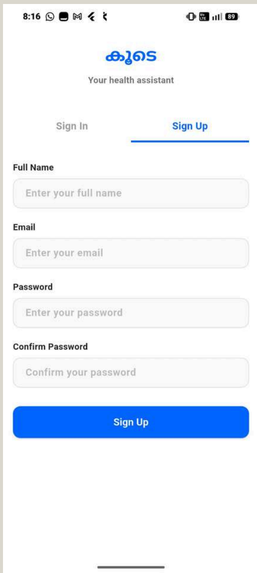


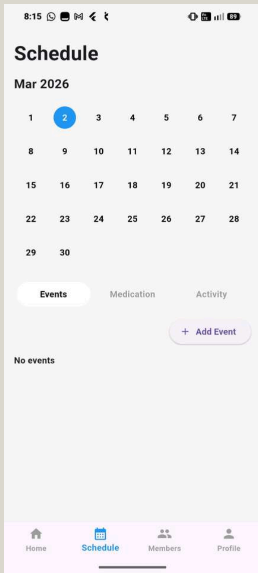
Patient returned Alert

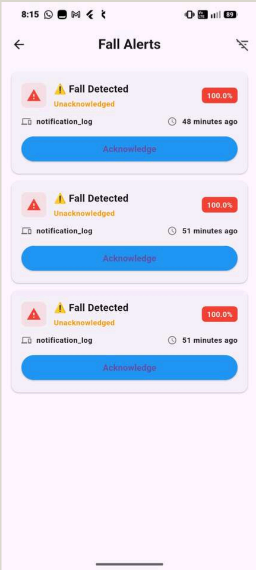


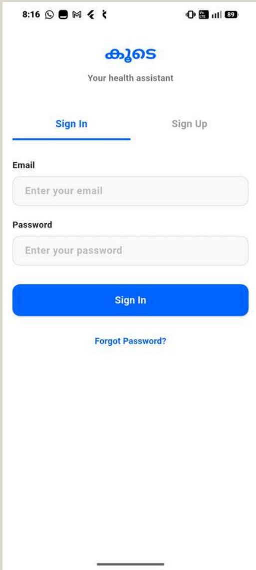
Fall Detection and Voice Recoginition Alert

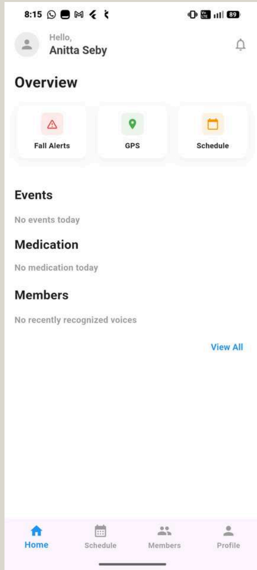


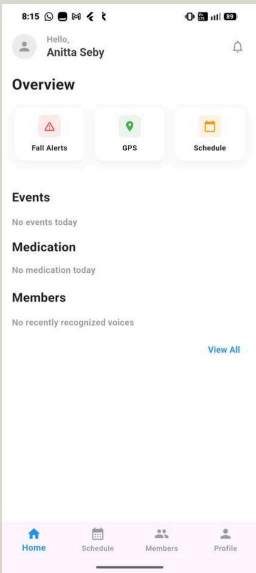





8:15
Fall Alerts
Fall Detected
Unacknowledged
100.0%
notification_log
48 minutes ago
Acknowledge
Fall Detected
Unacknowledged
100.0%
notification_log
51 minutes ago
Acknowledge
Fall Detected
Unacknowledged
100.0%
notification_log
51 minutes ago
Acknowledge


8:16
കൂടെ
Your health assistant
Sign In
Sign Up
Email
Enter your email
Password
Enter your password
Sign In
Forgot Password?


8:15
Hello,
Anitta Seby
Overview
Fall Alerts
GPS
Schedule
Events
No events today
Medication
No medication today
Members
No recently recognized voices
View All
Home
Schedule
Members
Profile


8:15
Hello,
Anitta Seby
Overview
Fall Alerts
GPS
Schedule
Events
No events today
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No medication today
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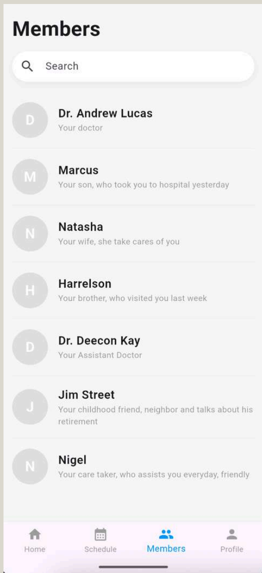
No recently recognized voices
View All
Home
Schedule
Members
Profile


Members
Search
Dr. Andrew Lucas
Your doctor
Marcus
Your son, who took you to hospital yesterday
Natasha
Your wife, she take cares of you
Harrelson
Your brother, who visited you last week
Dr. Deecon Kay
Your Assistant Doctor
Jim Street
Your childhood friend, neighbor and talks about his retirement
Nigel
Your care taker, who assists you everyday, friendly
Home
Schedule
Members
Profile

```
📊 All similarity scores:
AleenaP        → 0.796999990940094
Niranjana      → 0.7269999980926514
Angeline       → 0.6859999895095825
Maria          → 0.6850000023841858
Nitha          → 0.640999972820282
Aswathi        → 0.6230000257492065
DevikaAS       → 0.5889999866485596
Abhinav        → 0.5740000009536743
Mohan          → 0.5640000104904175
Ayush          → 0.5619999766349792
Sooraj         → 0.5440000295639038
Anatharam      → 0.5379999876022339
Athul          → 0.5099999904632568
Ashwin         → 0.4690000116825104

✅ FINAL RESULT:
🎯 Recognized Speaker: AleenaP
📈 Confidence Score: 0.796999990940094
```

```
Enrollment backend running on ws://0.0.0.0:8765
Client connected
[Anitta] Window 1/8
[Anitta] Window 2/8
[Anitta] Window 3/8
[Anitta] Window 4/8
[Anitta] Window 5/8
[Anitta] Window 6/8
[Anitta] Window 7/8
[Anitta] Window 8/8

✅ ENROLLMENT COMPLETE
Speaker : Anitta
Saved   : data\embeddings\Anitta.npy
```


ESP32 Code

```
#include <WiFi.h>
#include <ArduinoWebsockets.h>
#include <driver/i2s.h>

using namespace websockets;

/* ================= INMP441 I2S PINS ================= */
#define I2S_BCLK 26
#define I2S_LRCL 25
#define I2S_DIN  33
#define I2S_PORT I2S_NUM_0

/* ================= AUDIO CONFIG ================= */
#define SAMPLE_RATE        16000
#define I2S_READ_SAMPLES  512      // ~32 ms audio chunks

/* ================= WIFI ================= */
const char* ssid = "Test";
const char* password = "ashish26";

/* ================= WEBSOCKET ================= */
const char* WS_SERVER = "ws://172.20.10.4:8765";

/* ================= GLOBALS ================= */
WebsocketsClient ws;
QueueHandle_t audioQueue;

/* ================= I2S SETUP ================= */
void setupI2S() {
  i2s_config_t config = {
    .mode = (i2s_mode_t)(I2S_MODE_MASTER | I2S_MODE_RX),
    .sample_rate = SAMPLE_RATE,
    .bits_per_sample = I2S_BITS_PER_SAMPLE_16BIT,
    .channel_format = I2S_CHANNEL_FMT_ONLY_LEFT,
    .communication_format = I2S_COMM_FORMAT_I2S,
```

Backend Code

phase2_server.py

```python
import asyncio
import websockets
import numpy as np
import soundfile as sf
import os
from datetime import datetime
from resemblyzer import VoiceEncoder, preprocess_wav

# ================== CONFIG ==================
MODE = "enroll"                    # "enroll" | "idle"
CURRENT_SPEAKER = "Ayush"          # editable
WINDOW_SECONDS = 8
TARGET_WINDOWS = 10
SAVE_WAVS = True

SAMPLE_RATE = 16000
FRAME_SAMPLES = 512
TARGET_SAMPLES = WINDOW_SECONDS * SAMPLE_RATE

BASE_DIR = "data"
ENROLL_DIR = os.path.join(BASE_DIR, "enroll", CURRENT_SPEAKER)
EMBED_DIR = os.path.join(BASE_DIR, "embeddings")

os.makedirs(ENROLL_DIR, exist_ok=True)
os.makedirs(EMBED_DIR, exist_ok=True)

# ================== INIT ==================
encoder = VoiceEncoder()
audio_buffer = []
embeddings = []
window_count = 0

# ================== SILENCE TRIM ==================
```

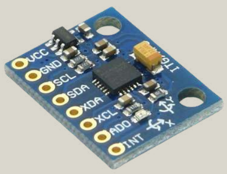
ASSUMPTIONS

- The patient consistently wears the IoT wristband during daily activities.
- The patient is in the mild or moderate stage of Alzheimer's and is physically mobile.
- GPS signals are sufficiently available for outdoor location tracking.
- Internet connectivity (Wi-Fi / Mobile Data) is available for real-time data transmission and alert delivery.
- The caregiver has access to a smartphone with the mobile application installed.
- Sensor data from the accelerometer and gyroscope is accurate and properly calibrated.
- The server remains operational for continuous processing and alert management.

Hardware Requirements

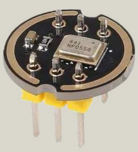
ESP32 Microcontroller

MPU6050 (Accelerometer & Gyroscope)



Neo 6M (GPS Module)

INMP 441 (Microphone Module)



Smartphone (Caregiver App)

Software Requirements

Backend

- Python (Flask Framework)
- Firebase Admin SDK
- Joblib (for ML model loading)
- NumPy

Machine Learning

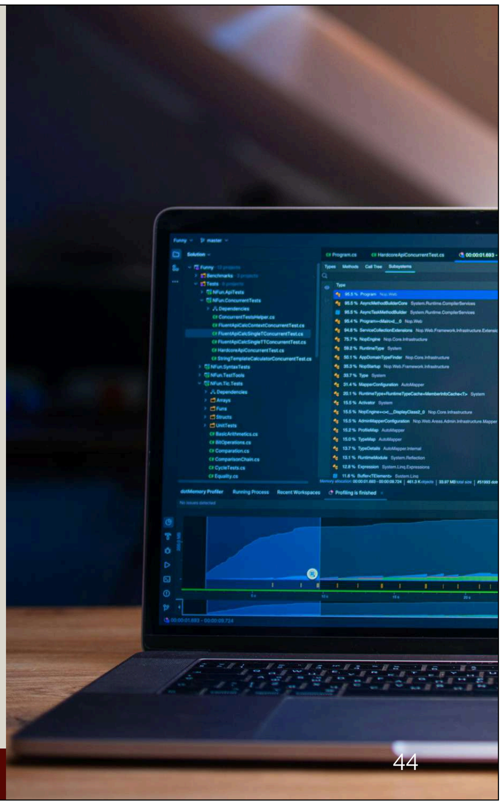
- Random Forest(Fall Detection Model)
- Model trained using accelerometer & gyroscope features

Mobile Application

- Flutter

Development Tools

- Arduino IDE
- VS Code



RISKS AND CHALLENGES

Risks:

- Patients may forget or refuse to wear the device.
- False alarms or missed alerts can cause stress or danger.
- Battery limits may stop continuous monitoring.
- Network failures can delay emergency alerts.

Challenges:

- Making voice recognition work well in noisy places.
- Getting stable GPS signals indoors or in remote areas.
- Designing the device to be comfortable and easy to wear daily.
- Keeping patient data secure and private with strong encryption.
- Lowering hardware costs while keeping good performance and reliability.

CONCLUSION

- The project develops a smart and secure IoT system to support Alzheimer's patients.
- It integrates real-time location tracking, fall detection, voice recognition, and cloud data management.
- Caregivers receive instant alerts and can monitor patients via a mobile app.
- The system addresses challenges like privacy, connectivity, and ease of use.
- It aims to improve patient safety, independence, and quality of life.
- The technology provides peace of mind for families and caregivers.
- Overall, it offers a practical, advanced solution tailored for Alzheimer's care.

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THANKYOU

Appendix B: Vision, Mission, Programme Outcomes and Course Outcomes

Vision, Mission, Programme Outcomes and Course Outcomes

Institute Vision

To evolve into a premier technological institution, moulding eminent professionals with creative minds, innovative ideas and sound practical skill, and to shape a future where technology works for the enrichment of mankind.

Institute Mission

To impart state-of-the-art knowledge to individuals in various technological disciplines and to inculcate in them a high degree of social consciousness and human values, thereby enabling them to face the challenges of life with courage and conviction.

Department Vision

To become a centre of excellence in Computer Science and Engineering, moulding professionals catering to the research and professional needs of national and international organizations.

Department Mission

To inspire and nurture students, with up-to-date knowledge in Computer Science and Engineering, ethics, team spirit, leadership abilities, innovation and creativity to come out with solutions meeting societal needs.

Programme Outcomes (PO)

Engineering Graduates will be able to:

1. Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems

2. Problem analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

3. Design/development of solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.

4. Conduct investigations of complex problems: Use research-based knowledge including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

5. Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.

6. The engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

7. Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.

8. Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

9. Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

10. Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

11. Life-long learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

Programme Specific Outcomes (PSO)

A graduate of the Computer Science and Engineering Program will demonstrate:

PSO1: Computer Science Specific Skills
The ability to identify, analyze and design solutions for complex engineering problems in multidisciplinary areas by understanding the core principles and concepts of computer science and thereby engage in national grand challenges.

PSO2: Programming and Software Development Skills
The ability to acquire programming efficiency by designing algorithms and applying standard practices in software project development to deliver quality software products meeting the demands of the industry.

PSO3: Professional Skills
The ability to apply the fundamentals of computer science in competitive research and to develop innovative products to meet the societal needs thereby evolving as an eminent researcher and entrepreneur.

Course Outcomes (CO)

Course Outcome 1: Model and solve real world problems by applying knowledge across domains (Cognitive knowledge level: Apply).

Course Outcome 2: Develop products, processes or technologies for sustainable and socially relevant applications (Cognitive knowledge level: Apply).

Course Outcome 3: Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks (Cognitive knowledge level: Apply).

Course Outcome 4: Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms (Cognitive knowledge level: Apply).

Course Outcome 5: Identify technology/research gaps and propose innovative/creative solutions (Cognitive knowledge level: Analyze).

Course Outcome 6: Organize and communicate technical and scientific findings effectively in written and oral forms (Cognitive knowledge level: Apply).

Appendix C: CO-PO-PSO Mapping

Course Outcomes

S.No	Description	Bloom's Taxonomy Level
1	Model and solve real world problems by applying knowledge across domains.	Apply
2	Develop products, processes or technologies for sustainable and socially relevant application.	Apply
3	Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks.	Apply
4	Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms.	Apply
5	Identify technology/research gaps and propose innovative/creative solutions.	Analyze
6	Organize and communicate technical and scientific findings effectively in written and oral forms.	Apply

CO-PO AND CO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	2	2	1		2	1	1	1	1	1	2	2	2
CO2	2	2	2		2	3	2	1	1	1	1	2	2	
CO3								3	2	2	1			3
CO4		1	2		3	2	3	2	2	3	1		2	2
CO5	2	3	3	1	2					1	3	3		
CO6			2			2	2	2	3	1	1			3

Justifications for CO-PO-PSO Mappings

Mapping	Level	Justification
101003/ CS822U.CO1-PO1	Medium	Applies engineering knowledge, mathematics, and computing fundamentals to model and solve real-world problems.
101003/ CS822U.CO1-PO2	Medium	Requires identification, formulation and analysis of engineering problems before developing solutions.
101003/ CS822U.CO1-PO3	Medium	Involves designing systems or solutions to address identified engineering needs.
101003/ CS822U.CO1-PO4	Low	Limited investigation or experimentation may be required to validate the developed models or solutions.
101003/ CS822U.CO1-PO6	Medium	Solutions developed may influence societal, environmental, or sustainability aspects of engineering applications.

Mapping	Level	Justification
101003/ CS822U.CO1-PO7	Medium	Encourages consideration of ethical and professional responsibilities while developing solutions.
101003/ CS822U.CO1-PO8	Low	Some level of teamwork may be required while working on modeling or project tasks.
101003/ CS822U.CO1-PO9	Low	Requires communication of results and findings through reports or presentations.
101003/ CS822U.CO1-PO10	Low	Involves basic planning and execution of tasks while implementing solutions.
101003/ CS822U.CO1-PO11	Low	Encourages independent learning and adapting new knowledge to solve real-world problems.
101003/ CS822U.CO1-PSO1	Medium	Uses core computer science principles to analyze problems and design appropriate technical solutions.
101003/ CS822U.CO1-PSO2	Medium	Involves algorithm design and application of software development practices to implement solutions.
101003/ CS822U.CO1-PSO3	Medium	Applies computer science fundamentals in research or innovative problem solving for societal needs.
101003/ CS822U.CO2-PO1	Medium	Utilizes engineering knowledge to design and develop products or systems.
101003/ CS822U.CO2-PO2	Medium	Requires analysis of user needs and technological feasibility before development.
101003/ CS822U.CO2-PO3	Medium	Focuses on design and development of solutions addressing real-world requirements.
101003/ CS822U.CO2-PO5	Medium	Requires use of modern engineering and IT tools for system or product development.

Mapping	Level	Justification
101003/ CS822U.CO2-PO6	High	Emphasizes sustainability and societal impact during solution development.
101003/ CS822U.CO2-PO7	Medium	Ethical and professional responsibilities are considered when developing socially relevant technologies.
101003/ CS822U.CO2-PO8	Low	Development activities may require collaboration among team members.
101003/ CS822U.CO2-PO9	Low	Involves communicating design ideas and development outcomes.
101003/ CS822U.CO2-PO10	Low	Requires basic planning and coordination during development tasks.
101003/ CS822U.CO2-PO11	Low	Encourages continuous learning of emerging technologies while building solutions.
101003/ CS822U.CO2-PSO1	Medium	Applies computer science fundamentals to design technically feasible solutions.
101003/ CS822U.CO2-PSO2	Medium	Uses programming and software development practices to implement solutions.
101003/ CS822U.CO3-PO8	High	Strongly relates to functioning effectively as a member or leader in multidisciplinary teams.
101003/ CS822U.CO3-PO9	Medium	Requires effective communication among team members during collaboration.
101003/ CS822U.CO3-PO10	Medium	Team leadership involves project coordination and task management.
101003/ CS822U.CO3-PO11	Low	Team interaction supports learning from peers and adapting new approaches.

Mapping	Level	Justification
101003/ CS822U.CO3-PSO3	High	Professional collaboration and teamwork support innovative computer science solutions.
101003/ CS822U.CO4-PO2	Low	Requires analysis and planning before executing project tasks.
101003/ CS822U.CO4-PO3	Medium	Involves designing and implementing project solutions based on planned activities.
101003/ CS822U.CO4-PO5	High	Requires effective use of modern tools and resources during project execution.
101003/ CS822U.CO4-PO6	Medium	Considers societal and environmental impact while implementing solutions.
101003/ CS822U.CO4-PO7	High	Emphasizes adherence to ethical practices and professional responsibilities.
101003/ CS822U.CO4-PO8	Medium	Task execution may involve teamwork and coordination among team members.
101003/ CS822U.CO4-PO9	Medium	Requires communication of progress, outcomes and documentation.
101003/ CS822U.CO4-PO10	High	Focuses on project management including scheduling, resource allocation and task execution.
101003/ CS822U.CO4-PO11	Low	Encourages adaptability and continuous learning during project implementation.
101003/ CS822U.CO4-PSO2	Medium	Uses software development practices to execute and manage project tasks.
101003/ CS822U.CO4-PSO3	Medium	Applies computer science knowledge in practical project development environments.

Mapping	**Level**	**Justification**
101003/ CS822U.CO5-PO1	Medium	Uses engineering knowledge to understand technological limitations and opportunities.
101003/ CS822U.CO5-PO2	High	Requires deep problem analysis and identification of research gaps.
101003/ CS822U.CO5-PO3	High	Proposes innovative design solutions to address identified gaps.
101003/ CS822U.CO5-PO4	Low	May involve limited investigation or validation of proposed ideas.
101003/ CS822U.CO5-PO5	Medium	Uses appropriate tools and techniques to analyze and develop innovative solutions.
101003/ CS822U.CO5-PO10	Low	Requires minimal planning while proposing new solution approaches.
101003/ CS822U.CO5-PO11	High	Strongly promotes independent learning and exploration of emerging technologies.
101003/ CS822U.CO5-PSO1	High	Applies computer science fundamentals to identify and solve complex technical problems.
101003/ CS822U.CO6-PO3	Medium	Requires presentation of design concepts and developed solutions.
101003/ CS822U.CO6-PO6	Medium	Communication may include discussion of societal or environmental implications.
101003/ CS822U.CO6-PO7	Medium	Requires responsible and ethical communication of technical results.
101003/ CS822U.CO6-PO8	Medium	Communication supports collaboration within team environments.

Mapping	**Level**	**Justification**
101003/ CS822U.CO6-PO9	High	Strongly relates to effective written and oral communication of technical findings.
101003/ CS822U.CO6-PO10	Low	Requires structured reporting and documentation in project management contexts.
101003/ CS822U.CO6-PO11	Low	Encourages continuous improvement in communication and presentation skills.
101003/ CS822U.CO6-PSO3	High	Communication of innovative computer science solutions and research outcomes.